

HISTORY OF CHANGES		
VERSION	PUBLICATION DATE	CHANGE
1.0	12/04/2024	Part B - Transferred content from proposal part B - Provided additional information about the costs indicated in section 3.2.1
2.0	27/05/2024	Part B - Added a table at Section 3.2 to map the expertise of each partner to their allocated tasks within the project. - Added a paragraph in Section 3.2 to acknowledge the complexity of patent research and analysis and identify the internal resources which will be mobilized to support the goal of filing 3 patents in the course of the project. - Added AVL as a replacement of PAN - Replaced all references to partner “PAN” with “AVL” - Updated figures in Section 3.2 to reflect the new proposed budget distribution Part A - Removed PAN from the consortium since they decided not to participate as part of the closing of this unit participating in the project - Added AVL as a replacement of PAN - Added one deliverable “D1.4: Analysis of the legal and ethical issues (AEGIS;R) [M12]” to report the output of “T1.5: Ethical principles and European regulations, fundamental rights, and social sciences” - Changed the title and description of “D1.3: Innovation management report (AEGIS;R)” to “D1.3: Innovation management and ethics oversight report (AEGIS;R) [M36]” to also accommodate the final results of T1.5 - Added D1.4 to the means of verification of Milestone 2: “Project scope codification” - Changed responsible partner for “D6.2: MVP (IMT;OTHER)”. IMT is now responsible for D6.2 to balance the contributions of the partners, since AEGIS had been assigned 4 deliverables already. - Changed the descriptions of T4.1, 4.3, 5.1, 5.2 to describe more sufficiently the planned work - Changed the descriptions of T3.2, T6.1, T6.3 and T6.4 to reflect the extra work that AVL and AVI will conduct in order to accommodate the tasks that PAN left out - Changed the list of participants for T1.1, T2.1, T2.2, T2.4, T2.5, T3.2, T4.1, T4.2, T4.3, T6.1, T6.3, and T6.4 to reflect that AVL will be involved in these tasks, accommodating the tasks that PAN left out.
3.0	12/08/2024	Part B - Removed references to SAP Part A - New resource distribution: ICCS takes over part of SAP’s work in WP1, WP3 & WP4; ISI takes over part of SAP’s work in WP2 & WP4; AEGIS takes over part of SAP’s work in WP2, WP3 & WP6. Other direct costs are distributed among those 3 partners to support the augmented activities

TABLE OF CONTENTS

1	EXCELLENCE	3
1.1	OBJECTIVES AND AMBITION	3
1.1.1	<i>CoEvolution Measurable objectives</i>	3
1.1.2	<i>Ambition.....</i>	7
1.2	METHODOLOGY	10
1.2.1	<i>Overall concept.....</i>	10
1.2.2	<i>CoEvolution Conceptual Architecture.....</i>	12
1.3	PILOTS AND USE CASES.....	24
1.3.1	<i>Pilot 1 Securing the AI Supply Chain of Manufacturing Infrastructure.....</i>	24
1.3.2	<i>Pilot 2 Securing the AI Supply Chain in the Autonomous Vehicles Domain.....</i>	25
1.3.3	<i>Synergies with relevant European and national Initiatives.....</i>	27
1.3.4	<i>Inter-disciplinary approach.....</i>	27
1.3.5	<i>Social Sciences and Humanities (SSH).....</i>	27
1.3.6	<i>Sex and Gender Analysis</i>	27
1.3.7	<i>Open Science.....</i>	28
1.3.8	<i>Research Data Management.....</i>	28
2	IMPACT	28
2.1	PROJECT’S PATHWAYS TOWARDS IMPACT	28
2.1.1	<i>Analysis of target stakeholders and expected benefits.....</i>	29
2.1.2	<i>Contribution to the outcomes of the topic and to wider impacts</i>	29
2.1.3	<i>Economic/Technological Outcomes.....</i>	31
2.1.4	<i>Societal Outcomes.....</i>	32
2.1.5	<i>Feedback to Policy Measures</i>	32
2.1.6	<i>Potential Barriers and Requirements</i>	32
2.2	MEASURES TO MAXIMISE IMPACT - DISSEMINATION, EXPLOITATION AND COMMUNICATION	33
2.2.1	<i>CoEvolution Dissemination plan.....</i>	33
2.2.2	<i>Communication plan.....</i>	34
2.2.3	<i>CoEvolution Exploitation plan</i>	35
2.2.4	<i>CoEvolution Intellectual Property Management Strategy.....</i>	36
2.3	SUMMARY	37
3	QUALITY AND EFFICIENCY OF THE IMPLEMENTATION	38
3.1	WORK PLAN AND RESOURCES	38
3.1.1	<i>Overall structure of the workplan.....</i>	38
3.1.2	<i>Gantt chart.....</i>	39
3.2	CAPACITY OF PARTICIPANTS AND CONSORTIUM AS A WHOLE	39
3.2.1	<i>Consortium as a whole and capacity of participants.....</i>	39
4	ETHICS SELF-ASSESSMENT.....	40
4.1	ETHICAL DIMENSION OF THE OBJECTIVES, METHODOLOGY AND LIKELY IMPACT	40
4.2	COMPLIANCE WITH ETHICAL PRINCIPLES AND RELEVANT LEGISLATIONS	41
5	REFERENCES.....	42

1 Excellence

1.1 Objectives and ambition

As the artificial intelligence (AI) market accelerates to over \$2575.16 billion by 2032, we're seeing a shift from siloed AI solutions to more collaborative systems (such as federated learning or peer-to-peer learning) and to off-the-shelf, pre-trained third-party AI models that are used as is, with no security guarantees. These interconnected and black-box-based AI solutions promise seamless data flows, models and services across different sectors, leading to optimized decision making and tailored responses to dynamic situations. However, the seamless integration and collaborative capabilities of AI systems introduce significant vulnerabilities. Adversarial attacks, data poisoning and man-in-the-middle attacks can compromise the integrity and functionality of these AIs. In addition, the AI lifecycle, from data collection to design to deployment, becomes vulnerable to these threats, making the entire AI ecosystem vulnerable.

As the domain of AI use evolves and embraces a more connected and collaborative paradigm, several unprecedented threats and challenges are emerging. At the heart of these threats is the increased risk posed by the integration of multiple AI systems. Such integration, while promising in terms of collaborative intelligence and enhanced functionality, introduces potential points of compromise. For example, evasion attacks become more insidious in this context. If one system in the network can be fooled by an adversarial input, it could inadvertently cascade misinformation to other systems in the network/collaboration. Similarly, the shared data ecosystem within these networked systems increases the risk of poisoning attacks, where malicious data introduced into one module can percolate and affect the integrity of others.

Apart from the above, additional vulnerabilities of AI models to adversarial attacks can be introduced when models are unaware of their environment. Without an understanding of the broader context in which they operate, AI systems rely solely on individual data points to make decisions, making them vulnerable to manipulation that exploits these isolated views. Without a comprehensive view of the operating environment, AI models may miss patterns or anomalies that only become apparent when the larger context is considered. Adversarial attacks, particularly those involving subtle input biases, can easily go undetected in such situations. Contextual awareness is essential to provide a holistic perspective that enables models to effectively distinguish genuine variations from adversarial intrusions.

Given the complex landscape of adversarial attacks described above, AI security experts seem to have too many options to consider. On the one hand, the widespread use of AI has created a boom in AI security where a constant stream of adversarial attack variations are provided, while on the other hand, defences are overly specialised to specific attacks, not formally verified and lacking in accountability. While there are some rudimentary frameworks with assessment of adversarial attacks and poisoning attacks as well as a set of defences, there is still no coherent security assessment flow on how to holistically identify AI security vulnerabilities and risks as well as provide coherent defences applicable to the entire AI design and development lifecycle regardless of the AI model paradigm in a supply chain. Google has just recently published the Secure AI Framework (SAIF), a conceptual framework for secure AI systems that correctly identifies the AI security issues as highly critical, especially in the AI supply chain. Finally, from a regulatory perspective, in order to harness the potential of AI without compromising security, there is a significant need for international standards that can guide the development, deployment, and continuous improvement of these AI systems. Such a framework would pave the way for more responsible, secure, robust, ethical and universally accepted AI practice, ensuring that the power of AI in a supply chain is harnessed without compromising security or ethics. To this end, after a multi-year process and several guidance documents, the EU and its member states are making intensive efforts to regulate AI through frameworks like the European Commission's AI Act, aimed at ensuring both safety and trustworthiness in AI applications across all industries. This regulation highlights the need for an AI paradigm shift, where each AI model will be accompanied by specific characteristics, risk values and formal descriptions that will enable transparency, security evaluation and trust.

1.1.1 CoEvolution Measurable objectives

To accomplish this vision, the project sets the following more detailed objectives:

Objective 1: Establish a comprehensive security and risk assessment process for the complete lifecycle of an AI model in a supply chain and beyond.
--

<i>Description: The complexity and interconnectivity of today's AI landscape necessitate a comprehensive framework that can secure both the AI supply chain and the entire lifecycle of AI development and deployment. While there exist many adversarial attack assessment and mitigation techniques there is no coherent end-to-end flow on identifying threats, vulnerabilities and associating them with specific risks. The European Union and its member states are making intensive efforts to regulate AI through frameworks like the European Commission's AI Act, aimed at ensuring both safety and trustworthiness in AI applications across all industries. Nonetheless, there appears to be a deficiency in a comprehensive system that can effectively manage the possible weaknesses and risks in a networked AI-supply chain. However, EU AI act dictates that risk levels should be associated with each AI highlighting the need of risk and security assessment flow. In CoEvolution, we aim to build a robust, adaptable</i>

risk analysis and security assessment framework that accounts for the evolving landscape of security threats and technological advancements of the AI as-a-target cybersecurity attackers. The framework acknowledges all the latest development in adversarial attacks for a broad range of AI paradigms (single models, collaborative learning architectures and interconnected AI agents/supply chain architectures) and provide a complete end-to-end assessment flow for the whole AI model lifecycle. **Related WPs: 1, 3, 4, 5, 6**

Challenge: C1.1: Identifying risk for heterogeneous AI components sourced from different vendors within the AI supply chain. C1.2: Assess holistically the security, robustness and trust, as well as the privacy standing of an AI model within a supply chain. C1.3: Attribute comprehensive security and robustness risk values that comply with the EU AI act concept C1.4: Lack of standardized way of quantifying an AI model security status against a broad spectrum of adversarial attacks	Innovation: • Evaluate the risks from a holistic perspective considering potential cyber security and AI related attack surface and relevant legislations (eg. EU AI act). • Collect risk from various sources including AI adversarial attack common knowledge base for AI-supply chain related risk , AI Risk Database, OWASP AI security, taxonomies for adversarial and machine learning attacks such as NIST AI 100-2 E2023 • Take into account cascading risks within the AI supply chain for use of third party AI models and/or Collaborative Learning (P2P or FL) as well as interconnected AI agents supply chain. • Provide a comprehensive security assessment engine that includes a data evaluation mechanism and an AI model fuzzer. • Propose and establish a series of AI model STR compliance metrics as a security assessment result in order to quantify an AI model security and trust through out its whole lifecycle
---	---

Work Program Compliance: “concerns have been raised over the security, robustness of the AI algorithms (including AI at the edge), including the risks of adversarial machine learning and data poisoning. Thus, it is important to promote security-compliant AI algorithms” also “Proposals should demonstrate awareness of the EU approach on Artificial Intelligence, such as the proposed Artificial Intelligence Act.”

Achievability/Verification - KPIs: **KPI1.1** Detection of at least 3 known vulnerabilities per AI model paradigm, **KPI1.2** Adversarial Attack Detection Rate / $\geq 90\%$, **KPI 1.3** Specification of at least 10 Security Assessment metrics for 3 categories of adversarial Attacks, **KPI 1.4** Specification of at least 3 risk levels on security and regulatory risk

Objective 2: Provide a Security, Trust and Robustness (STR) Defence Framework offering holistic AI model security-by-design approach throughout its overall lifecycle and under any AI realization paradigm (single, collaborative learning or interconnected AI agents supply chain).

Description: The Adversarial Attack defences research landscape while providing a plethora of defences for each attack type it can be considered disorganized since it doesn't provide a coherent, end-to-end, full AI lifecycle defence process that will introduce horizontal defences that are complementary from the design to the deployment phase of an AI Model. Also, there are several different AI realization paradigms apart from the traditional single AI model approach like Collaborative Learning (eg. P2P or FL) but also interconnected AI agents that interact in unison to a given problem and constitute a supply chain. In CoEvolution we offer a defence framework that rely on defence in depth approach for defending an AI model through its lifecycle regardless of the AI model paradigm. We offer defences for each phase of the AI model lifecycle (in its MLDevOp cycle) that are complementing each other. We also provide at deployment phase add-on gadgets that apart from enhancing further the security/robustness of a deployed model they also offer additional treats like security self-awareness (AI model adversarial attack detection and reporting), continuous update for resistance against new adversarial attacks and secure execution flows. We also offer a cybersecurity runtime monitoring engine that can collect and report possible adversarial attack incidents as well as manage the updating of the AI model defences as a mean of attack mitigation. **Related WPs:3, 4, 6**

Challenge: C2.1: Addressing in a coherent way the challenges of preventing adversarial attacks in collaborative AI (decentralised peer-to-peer, or federated learning) or AI agents interconnected (full supply chain) environments. C2.2: Provide unified, defence in	Innovation: • Introduce the concept of adversarial attack defence flows that are defences that start from the design phase of a model, are further enhanced during training/testing/validation and are also updated further during deployment as add-on gadgets. The flow provide defence in depth. Indicative robustification techniques used in these flows are Ensemble Learning defences, obfuscation and deep unrolling. • Protect against privacy-based attacks by introducing strategies like
--	---

depth adversarial attack protection approaches that can handle multiple adversarial attack types C2.3: Provide defences that operate collaboratively to defend the whole AI model lifecycle (design, train, deploy) C2.4: Monitor the AI model security at operation time and report adversarial attack incidents.	HE/SMPC or differential privacy. • Introduce add-on gadgets at AI model deployment phase that perform continuous update on defences, monitor the AI Model for inference attacks and report such incidents. • Provide a Security run-time monitoring system for inference adversarial attack on a deployed AI model within the AI supply chain. • Identify malicious clients or peers in a collaborative learning setting
Work Program Compliance: “Security-by-design concept and resilience to adversarial attacks” also “it is important to promote security-compliant AI algorithms, leading to possible certification schemes in the future.”	
Achievability/Verification - KPIs: KPI2.1 Design and develop at least 3 defence flow, KPI 2.2 Provide at least 4 CoEvolution gadgets KPI2.3 Effectively mitigate using at least one defence flow per AI model paradigm one or more Adversarial Attack categories KPI2.4 Establish at least 2 privacy enhancement strategies.	
Objective 3: Introduce and support a new paradigm on Security, Trust and Robustness for AI models compliant to the EU approach on AI and propose the AI Model Bills of Material as a universal AI Model description structure.	
Description: CoEvolution introduces a new paradigm on Security, Trust and Robustness for AI models based on the concept of Bills-of-Materials (BOM) that offers a unified, comprehensive approach on describing AI models in a supply chain and offer STR guarantees compliant to EU High-Level Expert Group on AI (AI HLEG) directives on trust and fairness, the European Data Governance ACT and the GDPR guidelines. The AI MBOM paradigm and the supporting CoEvolution framework will help facilitate the regulatory compliance to the EU approach on Artificial Intelligence such as the EU Artificial Intelligence Act. Related WPs:1, 3, 5, 6	
Challenge: C3.1: While there is a proliferation of AI models as open-source, in the market or in research, currently there is no standard or official mechanism with which an AI model is verifiable secure and assessed with specific guarantees, or a way to describe a model with enough details for potential users.	Innovation: CoEvolution will design and implement a novel AI MBOM mechanism, as well as provide a novel STR flow for defences and assurance mechanisms while providing telemetry and updating mechanisms to adjust to emerging threats and technological advancements, ensuring long-term resilience. The CoEvolution framework, will make sure that AI applications abide by moral standards and sector-specific laws. To safeguard sensitive data, it applies privacy-by-design principles and makes sure privacy concerns on data are addressed, HE is used for private data processing in FL schemes and access controls are applied when needed. To address ethical issues, it also encourages transparency and explainability-by-design using techniques such as Deep Unrolling. To this end CoEvolution has reserved a specific task, T1.5 to monitor the EU regulation in order for the project results to be harmonized with EU guidelines.
Work Program Compliance: “Establishing secure data sharing protocols, abiding by data privacy laws, and making sure that the information exchange complies with industry standards are all examples of compliance with this goal”	
Achievability/Verification - KPIs: KPI3.1 Design and Develop at least 2 variants of AI MBOM structures KPI3.2 Represent at least 1 AI model type (client-server, peer) using AI MBOM per AI model Paradigm KPI3.3 Provide at least 2 AI MBOM representation of an AI model for each Pilot use case	
Objective 4: Infuse into the overall AI model design operation continuum (design, training, deployment) context awareness mechanism that allow a deployed AI model to be aware of its constantly evolving cybersecurity environment and adapt to adversaries based on such a context.	
Description: CoEvolution envisions context-awareness as a key resilience and performance enabler. To achieve it, we need to facilitate relevant context-specific vectors into AI algorithms and models, that will internalise structural knowledge about the operating environment. To achieve effective adaptation, AI systems must possess awareness of the environment and its actors through cross-modal descriptors, enabling intelligent adjustments that aim to maximise a vector of rewards contingent on specified preferences. To this end, we need to be able to 1) employ basic sensing capabilities for uncertain context information, 2) interpret and classify context information, and 3) engage in reasoning to decide system policy adjustments based on contextual cues. The context vectors will be dynamically constructed, following a bio-inspired evolutionary approach, to reflect real-time environment conditions, avoiding hard-coded specifications. Related WPs: 4, 5, 6	
Challenge: C5: Tackling the constantly	Innovation: • CoEvolution will implement a dynamic adaptation strategy premised in

evolving cybersecurity environment landscape in collaborative AI settings (e.g., decentralized peer-to-peer and federated learning). This challenge is compounded by the dynamic temporal and spatial nature of adversarial attacks and cybersecurity threats, emphasizing the need for adaptive and time-sensitive solutions.	context awareness, by employing genetic algorithms to continuously evolve context vectors, to address emerging threats and attack vectors in heterogeneous networks, ensuring long-term resilience. • Additionally a decision support system will be designed, using Multi-objective Reinforcement Learning (MORL) or Decision Transformers (DT), that chooses the optimal mitigation action, considering the context vector and the candidate actions selected at the security assessment stage, as well as relevant KPIs. • Finally Bayesian techniques will be investigated to mitigate overconfidence and enhance calibration of the decision support system. Overconfidence is particularly problematic in safety-critical applications (or when making decisions in areas where limited data is available).
Work Program Compliance: “Inclusion of context-awareness in machine learning in order to boost resiliency.”	
Achievability/Verification - KPIs: KPI4.1 Identify and design least 3 Context specific vector KPI4.2 Demonstrate at least 2 AI model changes due to context variation per pilot.	
Objective 5: Provide the foundation and create the momentum for an STR AI supply chain ecosystem that offers to the AI community the CoEvolution hub of secure, trusted and robust AI models, trusted training datasets as well as the framework to utilize them in various AI paradigms (collaborative learning) and the AI supply chain.	
Description: By integrating the various components of the CoEvolution architecture we will create a complete end-to-end Security, Trust and Robustness assessment and enhancement solution. Eventually, we will provide context aware STR AI models that are characterized universally by their AI MBOM. Our objective is to create a one-stop hub that will provide to the AI community a fully operational and coherent STR risk assessment and security assurance flow that can be applied to a broad range of AI solution and is compliant with the MLDevOps as well as the EU AI regulatory framework. The flow will be complemented by a novel yet analytic ways of describing an AI model, the AI MBOM structure, its generation and management as well as mechanisms for AI model security monitoring and AI model context awareness management. Our objective is to set the foundation of a new STR paradigm that unifies the different research directions of STR AI research, it will take into account the AI supply chain in all its different perspectives and give to the AI community the means to follow such a paradigm. To achieve that we complement the CoEvolution paradigm through its hub with open source trusted datasets generated and/or assessed for their lack of poisoning and bias, with trusted CoEvolution developed AI models to be used by the AI community and with an AI MBOM management mechanism that includes an AI MBOM repository and an AI MBOM generation and management mechanism. Along with the provided CoEvolution framework for risk and security assessment and the STR defences tools, libraries, models and services we offer a full solution for the paradigm shift that allows enterprises, research organizations, regulatory bodies and certification entities as well as end users to adopt the approach and eventually build an ecosystem around it that will make AI solution more robust, secure and in line with the EU regulatory directives. Related WPs: 5,6	
Challenge: C5.1: A structured, comprehensive methodology flow on how to provide security assurance for AI models in a supply chain is missing C5.2: Provide trusted models and trusted data with security guarantees is not available in practice. C5.3: A globally acknowledged mechanism for security characterizing AI models for use by third parties is missing. Also, no security guarantees management mechanism is provided for AI C5.4: The AI security research and enterprise community is fragmented and there is no coherent way as providing secure MLDevOps	Innovation: • We provide a paradigm shift to existing Secure MLDevOps and AI model lifecycle where an AI model is risk and security assessed and then STR enhanced throughout its lifecycle while is uniformly characterized by its associated AI MBOM. • We propose an AI MBOM management mechanism and the accompanying tools and services • Coevolution offers a hub of tools, services, security assurance flow, AI MBOM management as well as a series of trusted pools (models, data) that will help the research community to build an ecosystem around structured STR MLDevOps • We empower the CoEvolution ecosystem establishment by offering to the AI community the means to comply with EU regulations
Achievability/Verification - KPIs: KPI5.1 Provide at least 4 Trusted AI models in the Trusted model pool, KPI 5.2 Provide at least 4 Trusted Dataset in the Trused Data Pool KPI5.3 Provide at least 3 tools and services to be shared as open access solutions through the CoEvolution hub.	

Objective 6: Integrate, validate and assess the CoEvolution overall solution in 2 complex pilots that feature all AI model realization paradigms (i.e. single, collaborative learning and Interconnected AI agents) showcasing the security of the overall AI supply chain.	
Description: The CoEvolution framework is going to be tested in two different pilots in order to address realistic problems of the industry where AI models are designed, trained and deployed. In the first pilot, led by Siemens, we validate the CoEvolution framework in a supply chain manufacturing infrastructure scenario that highlights the need for zero-trust architecture in localized AI models for security and trustworthiness that aims to support and augment the security and efficiency of a manufacturing plant hosting a number of AI models. In the second pilot, we are aiming to support the AI supply chain in the autonomous vehicles domain, providing guidelines for AI model designs, support the training process of Federated Learning between vehicles and drones as well as maintain and secure runtime operations with dynamic model updates and context awareness on the deployed models in the autonomous vehicles. Related WPs:2, 6	
Challenge: In a supply chain manufacturer scenario where multiple AI models may need to be deployed, securing, hardening and verifying models, is a challenging task, especially when AI models will become a commodity in the near future. Similarly in the autonomous vehicles where hardened AI models are crucial to public safety, AI models with context awareness will further enhance the robustification.	Innovation: The CoEvolution solution has been specifically designed to solve the challenges described in this objective. The project introduces the CoEvolution STR flow that is accompanying all AI models and can provide security guarantees to all involved parties. Also, the CoEvolution STR flow generates an alive document, the AI MBOM, that follows the AI model lifecycle and constantly gets updated so as to reflect the latest secure status of the AI model. Also, the CoEvolution through the introduced context awareness mechanism is also able to handle the multiple device IoT environment and its specificities.
Work Program Compliance: Security-by-design concept and resilience to adversarial attacks; Inclusion of context-awareness in machine learning in order to boost resiliency.	
Achievability/Verification - KPIs: KPI 6.1 – At least 4 use cases examined on the pilot trials. KPI 6.2 At least 2 AI models per pilot to be validated by CoEvolution KPI6.3 At least one AI Model full Supply Chain evaluated in the various pilot use cases.	
Objective 7: Provide a viable and well-structured business plan on the exploitation of the CoEvolution solution that will lead into a sustainable exploitable outcome after the duration of the project.	
Description: The project will demonstrate the financial benefits (including cost-effectiveness and efficiency gains) that its novel framework, with the provided guarantees from the STR flow and assurance mechanisms, to be introduced to the AI community with quantifiable evidence. Building upon this evidence based as well as on the intelligence gained through market research, we aim at making a concrete and compelling business case for the introduction of the CoEvolution framework to the market. To this end, sustainable business and financing models for the exploitation of the CoEvolution framework will be defined using the Business Model Canvas and lean value proposition design. These models will serve as the basis for elaborating and fine-tuning an effective business plan which will serve as the roadmap for the exploitation of the CoEvolution solution and the adaptation of the framework by the target AI community and linked market. In parallel, professional innovation management as well as the deployment of well-targeted dissemination activities throughout the course of the project will enable us to nurture a favourable, enabling environment for building through the CoEvolution framework the necessary momentum to a CoEvolution ecosystem and widespread market uptake. Related WPs:2,3,6	
Challenge: Currently the AI community has several tools and open-source solutions at its disposal to make a viable business case. However, what is very challenging is verifying and securing AI solutions, while providing verifiable and transparent guarantees in different access control layers per type of user.	Innovation: The CoEvolution solution provides unique features, such as the AI MBOM, the STR flow, the assurance process and the context-awareness capabilities that have the potential to make a difference in the relevant AI community/market. The solution is focused on security and offers security auditing/testing and most importantly assurance through generated security guarantees. While there are other security assurance frameworks none of them is focused on the specificities of the AI paradigm from the CoEvolution context.
Work Program Compliance: Security-by-design concept and resilience to adversarial attacks; Inclusion of context-awareness in machine learning in order to boost resiliency.	
Achievability/Verification - KPIs: KPI-7.1: Exploitation group established with at least 10 stake-holders. KPI-7.2: Business model canvas for 2 types of business and financing models. KPI 7.3 More than 3 exploitation plans and business plans for project sustainability.	

1.1.2 Ambition

1.1.2.1 Detection, Prevention and Mitigation of Adversarial Attacks on AI models

State of the Art: Data-driven artificial intelligence (AI) and machine learning (ML) techniques have shown to be successful in many different applications, including computer vision, speech recognition and cybersecurity-related domains like malware detection. Despite many commercial solutions relying on AI/ML algorithms have been deployed, such algorithms remain vulnerable to well-crafted attacks against them. The potential threats against AI/ML are well understood at the research level^{i,iii} and can be categorized depending on whether the attacker has the ability to influence the training data used to learn the AI/ML model, or only the test data that will be provided as input to the algorithm during operation. Attacks staged at training time are known as poisoning attacks, while attacks performed during operation include evasion and privacy attacks.

Poisoning attacks compromise the learning algorithm by injecting carefully crafted poisoning samples in its training data to cause either a denial of service (i.e., a substantial increase in the test error which makes the system unusable for legitimate users)^{2,iii} or specific intrusions/misclassifications at test time (e.g., via backdoors that can be activated at test time by specific input triggers)^{2,iv}.

Evasion attacks (a.k.a. adversarial examples) work instead against a trained model during operation. They consist of perturbing the input data at test time to have it misclassified (e.g., modifying digital input images to be misclassified by an AI/ML system)^{v,vi}. Finally, privacy attacks can extract confidential information either about the system itself or about its users by iteratively querying a machine-learning system, e.g., offered as an online service^{vii}.

Even though the research community has largely demonstrated that AI/ML algorithms are vulnerable to different attacks, a more thorough analysis of their practical impact and feasibility, together with proper testing tools to help AI/ML developers quantify the associated risks, is still lacking, especially when such models need to be continuously deployed and updated. The problem of mitigating such attacks from the AI/ML algorithmic viewpoint remains also very challenging, as they undermine the fundamental stationarity assumption of AI/ML algorithms, i.e., that training and testing data are drawn from the same distribution. Although the research community has produced many tools to optimize attacks against AI/ML models (e.g., FoolBox, Cleverhans, IBM ART, SecML), they require strong knowledge of the AI/ML security problems to be used correctly and do not provide accessible user interfaces. In addition, the implemented attacks and defenses are typically very computationally demanding, hindering their applicability in efficient industrial processes.

Beyond the state of the art: Within CoEvolution, we aim to overcome these issues and facilitate the deployment of more reliable, secure, and trustworthy AI/ML models in industrial/automotive software development processes. We will provide solutions to easily integrate existing ML testing libraries and defensive strategies to mitigate the impact of the aforementioned attacks into current software development lifecycles while making them more efficient and easier to configure and automate. We thus firmly believe that this project will bridge the gap between research in AI/ML security and its deployment in industrial processes, thereby improving the current best practices for deploying secure and robust AI models in cybersecurity applications.

1.1.2.2 Adversarial attacks on P2P and FL collaborative learning

State of the art: Traditional machine learning (ML) systems are based on a centralized design where all data is transferred, stored and processed in the cloud, and the user endpoints could later fetch the results via API calls. While this approach provides good performance from a ML perspective, it is extremely costly from a system point of view (i.e., storage, processing, bandwidth), when the cloud provider becomes also the single point failure. Further, it does not address successfully the significant privacy concerns, especially considering the sheer volume of the multidimensional and privacy-sensitive user data (e.g., the COVID-19 proximity tracking apps^{viii}) the modern mobile and IoT devices collect nowadays.

To mitigate these concerns, the distributed ML paradigm of Federated Learning (FL)^{ix} came to enable mobile devices to produce an ML model without sharing any user data. Instead, in FL, users selectively share subsets of their models' key parameters during training. However, FL requires the coordination of a central server that orchestrates the training, the client's selection, the updates collection, and weights aggregation. This server constitutes a centralized authority that (1) users are forced to blindly trust to protect them from various information leakage attacks^{x,xi}, while at the same time (2) such centralized service is a single point of failure, which can become a bottleneck impacting system scalability and performance^{xii}. To address these emerging issues, existing approaches either use (1) differential privacy^{xiii}, thus decreasing the utility of the shared gradients, or (2) specialized hardware (e.g., Trusted Execution Environments^{xiv}). Further, previous peer-to-peer learning systems, that do not require a centralized server, heavily rely on blockchain ledgers^{xv} and interplanetary file system (IPFS)^{xvi}, or again use differential privacy^{xvii}.

Beyond the state of the art: In CoEvolution we take the next step and propose P4L: a peer-to-peer learning system that does not rely either on differential privacy or on external systems (e.g., blockchain or IPFS). In particular, P4L is a privacy preserving, fully distributed and infrastructure-less peer-to-peer learning system that enables users to participate in an asynchronous decentralized collaborative learning scheme. With the proposed

collaborative learning framework the users will benefit from the wealth of data that are distributed across the participating devices, while maintaining accurate personalized models. Specifically, in P4L, users will be able to train their models locally by using their (privacy-sensitive) data and create asynchronous, small, collaborative learning synergies with nearby users, via which they will share their model's gradients. In P4L, we will utilize partial Homomorphic Encryption (HE) during gradient averaging to preserve both the confidentiality and utility of the shared gradients, while at the same time eliminate information leakage attacks. In addition, P4L will leverage the proximity and cross-device communication capabilities^{xxiii} of mobile devices, thus operating on completely infrastructureless setups: no need for any centralized federation, Public Key Infrastructure (PKI) or Internet connection. A truly open architecture for FL and, even more ambitiously, fully distributed P2PL, would enable devices and apps to collaboratively build better and more resilient models thanks to 1) larger and richer datasets and 2) leveraging contextual awareness, while safeguarding security and privacy.

1.1.2.3 Context-aware decision support for threat mitigation using multi-objective criteria

State of the art: Reinforcement learning (RL) has proven to be a successful method for tackling complex challenges, ranging from board and video games^{xix}, to robotics^{xx}, smart systems^{xxi}, and chip design/placement^{xxii}. In the traditional RL setting, a policy interacts with the environment to maximize a single cumulative reward (in model-based RL algorithms a simulator is employed to model the environment and the policy can learn by interactions with the simulated environment^{xxiii}). However, numerous real-world problems involve multiple, potentially conflicting objectives. For instance, in robotics tasks, there is a compromise between maximizing speed and energy consumption. Unlike single-objective scenarios, these environments are characterized by the measurement of performance using multiple objectives, resulting in multiple Pareto-optimal solutions based on objective preferences^{xxiv}. In these scenarios, multi-objective reinforcement learning (MORL) approaches^{xxv} aim to maximize a vector of rewards contingent on specified preferences.

Existing MORL algorithms can be broadly categorized into two main groups^{xxvi}, ∴ single-policy methods and multiple-policy methods. Single-policy methods focus on identifying the optimal policy based on specified preferences among the objectives^{xxvii}. These methods explore various forms of preference functions, including non-linear ones such as the minimum over all objectives or the count of objectives exceeding a certain threshold. However, single-policy methods face limitations when preferences are unknown. Multi-policy approaches target the approximate Pareto frontier of optimal solutions. Typically, multiple runs of a single-policy method over different preferences are generated^{xxviii}. Policy-based RL algorithms^{xxix} simultaneously learn the optimal manifold over a set of preferences. There are value-based RL algorithms that implement an extended version of the Bellman equation^{xxx}. Multi-objective fitted Q-iteration (MOFQI)^{xxxi} encapsulates preferences as input to a Q-function approximator and uses expanded historical trajectories to learn multiple policies. However, these methods explicitly maintain sets of policies, making it difficult to scale up to high-dimensional preference spaces. Also, these methods cannot be easily adapted to new preferences at test time.

Beyond the state of the art: We cast the context-aware decision support task as a recommendation problem. The role of the context-aware decision support system will be to choose the optimal mitigation action considering candidate actions selected at the threat intelligence stage and context vector. Concretely, we plan to implement a multi-objective decision support framework using Reinforcement Learning (RL) or Decision Transformer (DT) algorithms. In said framework, each action affects multiple KPIs of the system, and the changes can be mapped to positive and negative rewards, required to train a decision-making agent. The multi-objective dimension aims to integrate the utility functions of the individual objectives in place into a single reward function (e.g., weighted sum of the utility functions of throughput, latency, resource utilization, energy efficiency). As such, a central idea of our approach is to couple these rewards to target KPIs while accounting for context information.

On the one hand, relevant KPIs may include throughput (effective data rate of information successfully transmitted over the network), latency (time it takes for data packets to reach their destination), jitter (variance of a signal over time), packet loss (portion of packets that do not reach their destination in the network), resource utilization (the usage of network resources affects CapEx and OpEx) and energy efficiency (e.g. switch on/off devices based on the current network demand). On the other hand, we will consider contextual aspects (e.g., proximity, time, space) in the latent space of adversarial threats to deliver targeted policy augmentations into the adversarial training. These parameters cannot be treated independently, as this may produce sub-optimal solutions. More in general, we are dealing with a Pareto-optimal problem, where the goal is to find all Pareto efficient solutions, at which stage none of the objective functions can be improved in value without degrading some of the other objective values. However, we aim to develop an agent that can dynamically find the most relevant KPIs to maximize performance depending on the specific network state, by transitioning among the different Pareto efficient solutions. Finally, we plan to study Bayesian techniques to mitigate overconfidence and enhance calibration of the final algorithm. Overconfidence is particularly problematic in safety-critical applications (or when making decisions in areas where limited data is available), e.g. medical diagnosis, finance or mitigation actions on malicious activity on network systems. Regarding reinforcement learning algorithms, we plan to test and compare

the most relevant state-of-the-art ones. This includes deep Q-Networks and actor-critic approaches.

1.1.2.4 Technologies Introduced to CoEvolution and TRL Levels

Main Technologies	TRL	Indicative Details and Role in CoEvolution
TID P4L (Objective 2)	3	TID will develop a peer-to-peer learning approach that does not rely either on differential privacy or on external systems (e.g., blockchain or IPFS). In particular, said solution will consist of a privacy preserving, fully-distributed and infrastructure-less peer-to-peer learning system. Specifically, P4L will utilise partial Homomorphic Encryption (HE) during gradient averaging to preserve both the confidentiality and utility of the shared gradients, while at the same time eliminating information leakage attacks. In addition, TID will leverage the proximity and cross-device communication capabilities of mobile devices, thus operating on completely infrastructureless setups: no need for any centralised federation, Public Key Infrastructure (PKI) or Internet connection. Moreover, it will research context-aware extensions for enabling resilient and robust collaborative learning, along the research directions of CoEvolution. (End TRL 5)
AVISENSE CAV Simulation platform (Objective 5)	4	ATHENA CAV simulation platform is a comprehensive simulation platform specifically crafted for developing, validating, and testing Autonomous Driving software, with a special focus on scenarios involving constrained spaces. Its highly optimized and scalable architecture makes it an ideal choice for simulating CAVs, performing also radio propagation in V2X communication, based on ITS-G5 standard. (End TRL 6)
AVISENSE deep-unrolling based AI model designs (Objective 4)	3	AviSense brings its experience from the TRUSTEE project in modern model-based network design methodologies, transforming well-defined mathematical optimizations (stemming from the prior domain knowledge of the examined problem) into meaningful and interpretable deep learning architectures, known as deep unrolling (DU) networks. DU architectures organically incorporate deep regularizers for data quality enhancement, which we will utilize for resilience against adversarial attacks, enriching the design-phase based tools and services of the CoEvolution ecosystem. (End TRL 5)
ISI Side Channel Leakage Assessment Platform (Objective 1)	3	FlecLeCo v2 is a Side Channel leakage collection and analysis platform for hardware and software implementation power consumption and electromagnetic emission leakage trace acquisition and analysis. The platform performs security assesment on Side Channel Attacks (SCAs) using traditional SCA techniques but also ML/DL profiling techniques. During CoEvolution the platform will be used in order to assess AI models against Side Channel leakage that can be exploited for Model extraction/stealing and privacy leakage. (End TRL 5)
ISI Scalable HE /SMPC primitives library for AI schemes (Objective 4)	3	ISI provides a HE/SMPC add-on module build on top popular open source HE/SMPC libraries like TFHE/concrete that offers scalability in order to be efficiently applied to the AI supply chain paradigm (End TRL 4)
PlanIoT (IMT) (Objective 4)	3	PlanIoT enables the management of sensor data in an optimized manner based on the application requirements, context information and available resources. In CoEvolution, PlanIoT will be extended to support context-aware AI resilience by dynamically updating the context of IoT data and resources in distributed FL and decentralised P2PL architectures. (End TRL 5)
UNICA Secure Machine Learning (SecML) Library (Objective 6)	3-4	UNICA provides SecML, an open-source Python library for AI testing and adversarial robustness enhancement in different application domains, considering both training-time and test-time adversarial attacks (End TRL 6-7)

1.2 Methodology

1.2.1 Overall concept

The CoEvolution is envisioned as a multilayer hub of building security and trust into Artificial Intelligence solutions by offering robustification tools and services, security assessment and assurance mechanisms, trusted AI models and dataset pools as well as capacity building mechanisms that instil security, robustness and trustworthiness to AI models from three different viewpoints, the security of the whole AI lifecycle, security of the full AI ecosystem (Individual

AI models, Collaborative Learning (Federated Learning, decentralized Learning etc.) paradigm models, as well as interconnected AI model agents in an AI supply chain) and the provision of security assurance of all AI model types. **From the AI lifecycle viewpoint**, CoEvolution provides security enhancement and robustification in all the steps of the AI model lifecycle. This includes security measures to be included for the model design phase, the model training/testing/validation phase (including mechanisms to evaluate the associated data) as well as the deployment and operation phase of an AI model (when performing inference on a given setting). All defences included in the AI model will be part of some Defence Flow, a continuous approach from the design to the deployment/operation phase, complementary to each other based on the security target goals. Eventually, using the CoEvolution, we envision the whole AI model lifecycle to be approached through a security-by-design, privacy-by-design and explainability-by-design perspective that will result into AI models with enough associated security assurance guarantees that render them trusted. Through the CoEvolution approach, we also infuse an AI model through its whole lifecycle with context awareness so that it can respond differently based on the environment on which it is deployed but also introduce techniques that allow an AI model to monitor its security status (detect when it is attacked) and report it to the CoEvolution security runtime monitoring system.

From the AI supply chain viewpoint, CoEvolution acknowledges the AI ecosystem diversity where AI models are been designed and used in various ways. Thus, in CoEvolution we consider AI adversarial attacks (and providing resistance against them) for individual, single, AI models that operate autonomously in a given system but also for collaborative AI model approaches that operate under a Federated Learning (FL) or a distributed, decentralized, Peer-to-Peer Learning (P2P). Finally, CoEvolution also provides the mean to secure the whole AI supply chain that may include interconnection of AI models, (interconnected AI agents) that cooperate collectively by exchanging inference results towards a given cause.

From the AI Security and Trust Assurance viewpoint, CoEvolution aims to provide AI models through its security enhancements and robustification operation flows (the CoEvolution security assurance flow) that will be considered secure/trusted. To achieve that in CoEvolution we provide a Security Assurance mechanism that records all the assessment and robustification activities that an AI model and associated data undergoes in order to become secure/trusted and generates a security credential that accompanies the AI model through its lifecycle. The CoEvolution security assurance flow also provides a management mechanism for the AI model credentials that includes apart from credential generation, also credential update as well as revocation.

The above three viewpoints in CoEvolution are supported by a universally used structure that is associated to an AI model, the AI Model Bill of Materials (MBOM). Adopting the logic of the Software Bills of Materials (BOM) accompanying software solutions, the MBOM provides detailed information about an AI model including the dataset used to train it, the owner and users, its dependencies to ML/DL libraries etc. but most importantly it incorporates all security related information associated with the use of the CoEvolution solution. The AI MBOM acts as the vessel of the CoEvolution security assurance generated credentials and follows an AI models lifecycle. It is managed by CoEvolution in an effort to create the necessary momentum for its broad adoption by the AI community as a mechanism of reporting robustness, security, trust, explainable and fairness of an AI model.

The CoEvolution solution is supported by a novel context awareness mechanism that relies on a multi-objective decision support system that is applied on top of any collaborative learning protocol for training AI models. The proposed framework is powered by a context awareness centralized and continuous adaptation engine based on bio-inspired evolutionary strategies observed in animal species. In general, as shown in Figure 3, context awareness is separated into three phases: (1) sensation, collecting information via sensors, (2) perception, identify context via collected information, (3) awareness and action, perform update and change behaviour based on context.

To achieve such adaptation, AI models in CoEvolution are equipped with a unique class of awareness – awareness of the environment and its actors via the provision of an embedding space of cross-modal descriptors - and the capacity to react and interact with its environment, via intelligent adaptations that optimise relevant reward functions. Both security-by-design and context awareness features of our solution lead us to the long-term goal of CoEvolution acting as an enabler to a concrete ecosystem around security and trust in AI solutions that captivates the different aspects of ML/DL research and development through their whole lifecycle. To achieve our goal, CoEvolution allows the exchange of AI security information using our novel AI MBOM approach, provides the support for exchange of trusted models and datasets (with security guarantees) and deliver the tools and methods to assess, harden, assure and monitor from a security perspective (under a series of adversarial threat models), via the Defence Flows and context awareness, for deployed AI models in popular use cases.

1.2.1.1 CoEvolution Considered Security Threat Models

In CoEvolution, we are considering all three prominent types of adversarial attacks currently recognized by the AI developer community, **black-box attacks**, when attackers have no existing knowledge of the AI model and rely on trial and error to manipulate systems, **white-box attacks** involve cybercriminals who have complete knowledge of the AI model or machine learning algorithm from its weights to its architecture and **gray-box attacks** that involve an attacker that has at least a partial amount of information and knowledge about the model's inner workings.

However, to achieve a concrete focus on the attack scope we apply the above categorization on specific *threat models (TMs)*. A threat model lists a set of assumptions related to the capabilities of an attacker and their goals of exploiting ML concepts for degrading the security and privacy of a system and its users. In what follows we provide brief description of such TMs: **TM1: ML model replacement**, **TM2: Poisoning of training data**, **TM3: Adversarial examples**, **TM4: Membership inference, model extraction and model inversion**.

1.2.2 CoEvolution Conceptual Architecture

The above-described concept of the CoEvolution hub, its capabilities and its viewpoints is manifested in the architecture that is presented in Figure 1.

The AI Risk Assessment and Guidelines/Recommendations Engine, drawing from a cybersecurity threat intelligence backbone provides for a given AI model design AI-associated risks related to the model's chosen design structure, employed type, training/testing/validation as well as its deployment in an AI supply chain (Federated Learning, Decentralized/Distributed/Peer2Peer, or interconnected AI agent approach). Based on the discovered risks the engine provides actionable insights on risk mitigation associating each risk with recommended mitigation techniques and tools from the CoEvolution Security Trust and Robustness Defense Framework. In association with the above risk engine, the Security Testing/Assessment engine discovers existing vulnerabilities on a given AI model design (to be trained or pre-trained) as well as data bias that can be used for data poisoning attacks on datasets that are used for training/testing/validation. Similarly, to the AI Risk Assessment and Guidelines/Recommendations Engine, the Security Assessment Engine offers recommended AI model robustification/hardening techniques and tools from the CoEvolution Security Hardening Toolbox to mitigate the discovered threats.

The above two engines draw information from existing open source knowledge bases on AI supply chain related risk and/or adversary attacks/vulnerabilities like the AI Risk Database and MITRE ATLAS. Eventually, the engines upon discovering new vulnerabilities on existing, publicly available, AI models they are capable of report such vulnerabilities to such knowledge bases. It should be mentioned that the above two engines produce reports that are included in the MBOM of an assessed AI model.

The Core of CoEvolution is the Security Trust and Robustness (STR) Defence Framework that consists of methods, libraries, trained models, tools and services capable of increasing the resilience/robustness of an AI model as well as removing data noise or bias on AI training/testing/validation data in all phases of an AI model/data lifecycle. The Framework solutions are split into 3 categories, the design phase, the training phase and the deployment phase libraries/tools. For each phase we are providing solutions that consider the various AI design paradigms including individual AI model robustification and adversarial attack prevention, detection and mitigation, Collaborative AI robustification (FL and P2P Learning) and also Interconnected AI model agents that exchange inference results between them. The component is further analysed in subsection 1.2.2.4. Eventually, when an AI model is designed, trained and deployed in an actual system following the CoEvolution Security assurance flow (supported by the STR Defence Framework) it integrates one or many CoEvolution defence flows into its structure. A defence flow is envisioned as a series of complementary defence techniques on each AI lifecycle phase and across phases in the AI Model robustification process that adopt a common AI core technology (eg. Ensemble Learning). At deployment phase the defence flow is realized as several "gadgets" that are added to the AI mode structure and they activate the AI model's robustness self-awareness and context self-awareness. These CoEvolution gadgets allow the robust CoEvolution enabled AI model to detect inference related adversarial attacks (eg. poisoning attacks) and report it to the CoEvolution Security Runtime Monitoring Engine but also to adapt it-self to the context that it operates on dynamically. Hence, the STR Defence Framework subcomponents (mapping the CoEvolution 3 lifecycle phases) will include proper context-awareness mechanisms that will eventually be infused (at the deployment phase tools and services) into the final robust AI model.

The CoEvolution context-awareness approach is diversified based on the architecture paradigm (single, collaborative, Interconnected) followed for an AI model. It can follow a static ruleset of context diversification (for single AI models) or a dynamic, evolution-based context adaptation (for the collaborative and interconnected paradigm) that involves dedicated "gadgets" based on reinforcement learning in the CoEvolution robust AI Model as well as a Central Context-Awareness Engine that collects context input vectors and using evolutionary algorithms generates new, more relevant to the condition at hand, context vectors. In CoEvolution, as context, we acknowledge information related to adversarial attacks on a given time and space as well as information on any use-case in-field (e.g., Sensors, network etc.) collected data (and metadata) of the system where an AI model is deployed to.

As CoEvolution robust AI model driven applications go live, the deployment phase “gadgets” for the models’ security-awareness, infused by the deployment phase solutions of some defence flow, take effect and provide feedback to the AI Security Runtime Monitoring Engine (SRME), offering adversarial cyberattack surveillance of AI operations. The goal of this engine is to provide to an operator the means to respond to a possible adversarial attack focused on the run-time operation of an AI model. Such inference focused

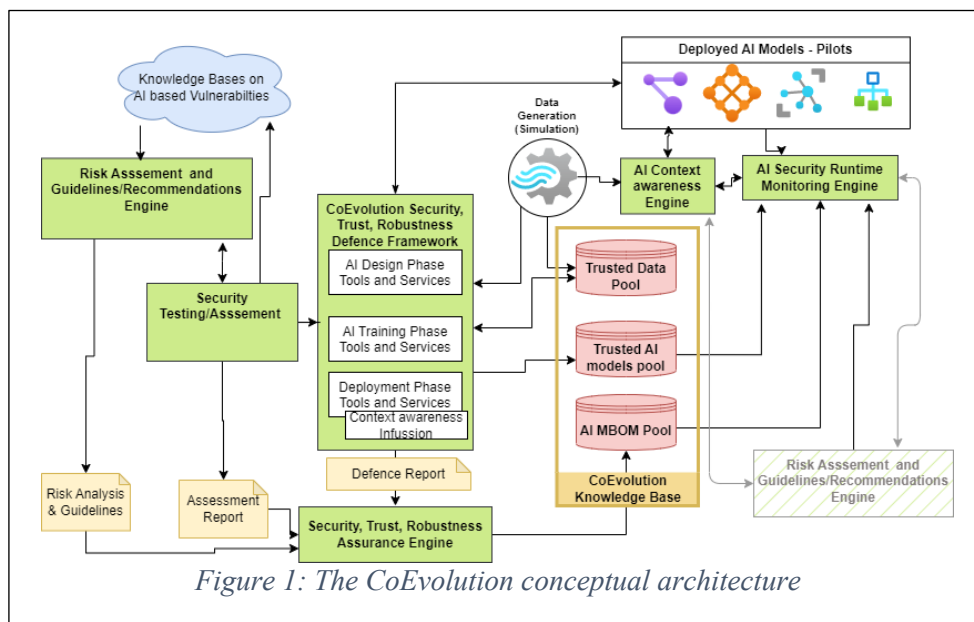


Figure 1: The CoEvolution conceptual architecture

attacks can be characterized as anomalies by the AI models and trigger an event to the SRME that will then inform its users and also provide updates to the MBOM associated with the attacked AI model. This engine is also providing a trust evaluation of any AI model associated with the CoEvolution hub. Through its association with the AI MBOM pool where all the generated/managed AI MBOMs are stored, the SRME is acting as a AI root of trust able to answer questions related to the validity, integrity and authentication of existing AI MBOMs associated with an AI model and with the update or revocation of an AI MBOM when an AI model is marked as compromised (based on information collected from the deployed AI model gadgets).

As expected, the CoEvolution hub supports a knowledge base with three data pools. The first data pool includes trusted AI models that have passed through the security scrutiny process of CoEvolution and include all the CoEvolution robustness, security, trust and context awareness features. These models are offered under an open access license (they may also rely on public AI models) to the AI community and aim at enhancing the CoEvolution ecosystem. Similarly, CoEvolution hosts a trusted Data pool that include open access trusted data that have been security assessed using the CoEvolution solution or they have synthetically been generated in a trusted manner using the CoEvolution Simulator component. Finally, the third data pool of the CoEvolution knowledge base is related to the management of the AI MBOMs that have been generated by the CoEvolution framework. This pool is accessed by the CoEvolution security assurance engine that generates new AI MBOMs for any CoEvolution secure/trusted, robust, context aware AI model in order to store such MBOM but it is also accessed by the CoEvolution SRME that updates, verifies or revokes an AI MBOM based on the operational status of the associated AI model.

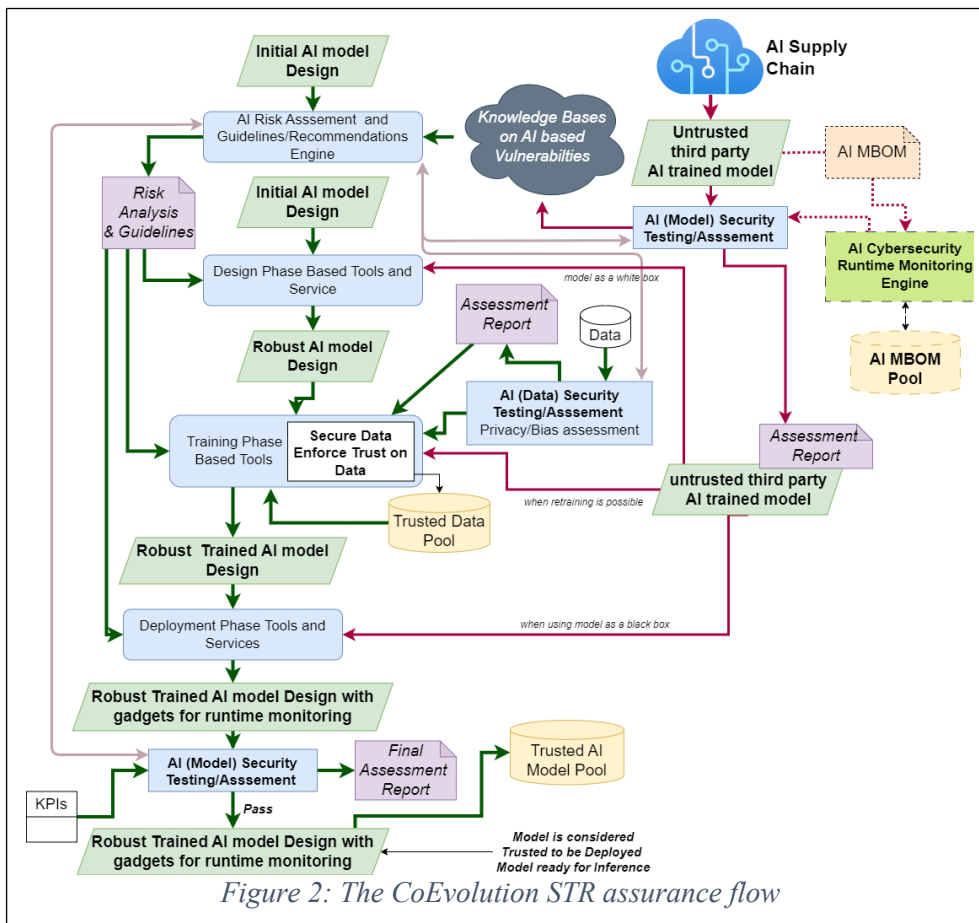
The CoEvolution Security Assurance Process acts as an orchestration component of the overall CoEvolution Risk and Security assessment, and Security robustness/context awareness process of the CoEvolution hub. This Process coordinates the above activities, aligns them with the guidelines and standards on AI trustworthiness and fairness, collects the various other engines reports and generates based on those reports the MBOM for a given assessed/secured/robust AI model.

1.2.2.1 AI Model Security, Trust and Robustness Assurance Flow

The main functionality of the CoEvolution is to provide Security assessment, security enhancement and overall security assurance with trust guarantees for the full lifecycle of AI models in an AI supply chain (regardless of the employed model paradigm: single, collaborative or interconnected). This functionality is supported by the Security, Trust and Robustness (STR) assurance flow of the CoEvolution project. The STR assurance flow is presented in Figure 2.

Initially in the STR flow, the CoEvolution user provides an initial draft design of an AI model to the CoEvolution risk assessment guidelines and recommendation engine. The engine is able to associate Risks with the various design components of the AI model (eg. each model layers or activation function etc.), risks related to the data to be used for training as well as security risk associated with the model as a whole. The outcome of the security risk analysis

is a risk analysis and guidelines report that apart from risk assessment it also provides guidelines and recommendations on which security enhancement tools and techniques to use to mitigate the risk. The engine draws intelligence from the knowledge accumulated from publicly available knowledge bases of AI adversarial attack techniques and related AI model vulnerabilities. The next step in the STR assurance flow is associated with the realization by the CoEvolution user of an STR enhancement plan starting from the initial AI model design, that utilizes sequentially components of the Security enhancement engine matching each AI model lifecycle phase. Thus, initially, the design phase toolbox of CoEvolution is used on the initial AI model



design aiming to modify the AI model structure so as to include proper adversarial attack countermeasures that can ensure this model remains robust and trusted under a hostile environment. However, in CoEvolution we do not consider design phase countermeasures enough to withstand attacks from strong adversaries (based on the identified CoEvolution threat models). A significant amount of adversarial attacks are related to how a model is trained and the data that are used for the training process. So, in parallel to the design phase security enhancement, CoEvolution utilizes the security assessment engine for evaluating the training data of an AI model. Utilizing this engine the data that are meant for training a model are assessed for bias, for fairness or for being poisoned by an adversary. If malicious data are discovered these data are removed from the data set and/or are replaced with synthetic values from simulation. It should be mentioned that CoEvolution adopts a secure training process as part of the training phase tools and services component as well as a secure mechanism for data cleaning, feature extraction and the feature selection mechanism that are typically considered during an AI model's training flow. Note that data, using the CoEvolution approach after passing the security enhancement process, can be stored in the trusted data pool of the CoEvolution knowledge base. It should be mentioned that apart from using data that are out of the control of CoEvolution (i.e originating from third-parties), during training the CoEvolution hub offers a use-case specific Data Generation Mechanism (The CoEvolution simulator) that can generate data considered trusted. Such data can act as ground truth for third party relevant datasets, but also they can be used at specific CoEvolution tools for the training process. In addition to the above, the Date Generation Mechanism can be used for testing the CoEvolution generated models in a simulation environment.

After the AI model has been trained using the training phase-based tool the outcome will be a robust trained model with proper robustness design to withstand adversarial attacks. However, given that such AI model will be deployed on a hostile environment where inference adversarial attacks may be mounted, CoEvolution also introduces a deployment phase toolset for securing an AI model during its operation phase. These tools are utilized in order to introduce within the AI model modules that do not necessarily influence the structure of the AI model itself but rather support the security of the AI model when it is deployed in a actual system. Depending on the AI paradigm followed by the AI model (single, collaborative or interconnected) such deployment mechanisms may differ yet they include

techniques that can guarantee a model or its outcomes are not tampered with, they do not succumb to inference adversarial attacks, they do not become non responsive etc. A key aspects in all the employed deployment techniques is the introduction of “gadgets” i.e additional modules within the AI model structure that are providing security and context self-awareness to the model itself. Apart from the above, the model will utilize such gadgets, introduced during the CoEvolution deployment phase security enhancement, to report attack attempts to the CoEvolution Cybersecurity runtime monitoring engine. Thus, the outcome of the Deployment Phase security Enhancement (using the Deployment Phase Tools and Services) is a robust trained AI model with gadgets that can prevent, detect and mitigate adversarial attacks on the wild.

However, this step doesn’t constitute the final act in the CoEvolution Security Enhancement flow. The outcome of the Deployment Phase security Enhancement is fed to the CoEvolution Security Testing/Assessment engine for a final security assessment so as to identify possible emerging vulnerabilities during the enhancement process and also to verify that the chosen tools were utilized properly to robustify the AI model. Eventually, this final security assessment produces an Assessment report that is used by the Security Assurance engine in order to associate the final generated AI model with the associated AI MBOM.

It should be noted that apart from the main STR assurance flow, the approach supports the security assessment and enhancement of third-party AI models (designs as well as trained models) complying with the concept of supply chain AI approaches (using third party models to enhance AI applications). In this case, the third party model is given as input to the CoEvolution Security, Trust and Robustness (STR) Testing/Assessment engine where it is evaluated for its security and robustness. This engine provides a security assessment report that includes also guidelines/recommendations on which tools and services of the STR enhancement engine to be used to create a CoEvolution compliant robust AI model with gadgets. Depending on third party model type, i.e. if the model is closed (black box) and pretrained (the most usual case), if it can support retraining or if it is described as an open AI model source code (where the hyperparameters, bias etc. as well as the model structure is known), the STR assurance flow then continues at the proper phase of the STR enhancement as described in the previous paragraphs. If the third-party model has a CoEvolution compatible AI MBOM then this is considered in the security assessment engine after the AI MBOM validation using the CoEvolution Security Runtime Monitoring engine AI MBOM management flow.

The final step in the STR assurance flow is the generation through the CoEvolution Security Assurance Engine, of the AI MBOM structure. Considering that this Engine orchestrates the STR assurance flow, it collects all the reports from the various other engines of the flow and extracts a structured history of the AI model security vulnerabilities/risks and the techniques/tools that were used for STR enhancement as well as additional information that uniquely characterize an AI Model (eg. dependencies, used ML/DL libraries etc) and generates an AI MBOM associated with the final robust AI model with deployment gadgets. Such AI MBOM is stored in the CoEvolution AI MBOM pool and is managed by the CoEvolution Security Runtime monitoring Engine.

1.2.2.2 Risk Assessment, Guidelines and Recommendations

The CoEvolution risk assessment, guidelines and recommendation engine (RAGRE) aims to provide an effective risk management practice to secure the overall AI supply chain ecosystem and the entire lifecycle of AI development and deployment. It provides a **novel way to evaluate the risks from a holistic perspective** considering potential cyber security and AI related attack surface and relevant legislations. In particular, the risks will be identified based on the security attacks generated from the individual heterogenous AI Models and their dependencies with others within the supply chain and unique utility-based and privacy-based AI adversarial attacks including data poisoning, adversarial and evasion attacks, backdoor attacks, model inversion, model extraction/theft, and many more. Hence, AI based risk identification aligns with AI audit requirements and considers AI system's purpose, context, and human roles, upholding transparency and stakeholder engagement by ensuring clarity in AI model methodologies, while rigorously assessing biases and impacts against protected categories. Additionally, the component will investigate the emergent threats and risks that pose the possible violations of the relevant legislation including **GDPR, Digital Services Act (DSA) and proposed EU AI Act** in order to tackle **data privacy and ethical consideration** and ensure **clear and transparent view of AI models' decisions** and appropriate data usage for ethical and responsible AI development.

RAGRE is **comprehensive and provides a systematic way** to assess and manage the risks proactively and adopts existing common knowledge bases to support the activities. It **extends the existing risk management practice** by incorporating assessment of **cascading risk** based on the attack path generated from multiple dependent AI components and relating vulnerability chain. The RAGRE AI security related risk assessment considers common knowledge base including AI-supply chain related risk, AI Risk Database, OWASP AI security, taxonomies for adversarial and machine learning attacks such as NIST AI 100-2 E2023. RAGRE also performs cybersecurity risk assessment in an effort to consider the context (i.e. the processing, network and infrastructure environment) in which an AI model is designed and implemented to operate. Thus, RAGRE extracts cybersecurity risks by considering information from existing knowledge bases including common vulnerability database, CVSS 4.0 scoring, CAPEC platform enumeration, ISO27001 and others. It should be mentioned that RAGRE operates in collaboration with the

Security Trust and Robustness Testing/Assessment Engine (described in detail in the following subsection). The output of this engine that performs thorough security testing/assessment for AI models and their data, are considered as evidence for RAGRE to extract security risks and using them to extrapolate realistic context aware regulation/legislation risks (in line with the proposed EU AI act). From that perspective, RAGRE creates a regulatory sandbox for an AI model under assessment and considers the relevant EU AI regulations, legislations, standards and policies apart from security testing evidence.

The **novelty of the CoEvolution RAGRE** is based on the correlation between cyber security and AI attacks based on the individual AI components and their dependencies to evaluate the **individual and cascading risk**. The identified risks in RAGRE are evaluated by following a Technical Evaluation Engine within a regulatory sandbox that scrutinizes AI models across multiple risk verticals, ensuring they meet stringent regulatory and industry standards. It employs continuous monitoring through the STR Testing/Assessment Engine and the security analysis from cybersecurity knowledge bases to detect and address performance drifts and utilizes adversarial attacks and stress tests to reinforce model defenses and estimate the risk level. The CoEvolution RAGRE is also aligned with the AI MBOM structure and its capabilities and provides direct input to it. The outcome of this engine is to ensure that individual AI models within a supply chain and/or AI systems based on collaborative learning are **reliable and safe thus promoting a more secure, trusted and robust AI supply chain eco system**.

1.2.2.3 Security, Trust and Robustness (STR) Testing/Assessment

The CoEvolution STR Testing/Assessment Engine is a direct extension of the Risk Assessment, Guidelines and Recommendations Engine (RAGRE). It utilizes the the initial risk assessment results to perform adversarial attack security testing on the design, training and deployment phase of an AI model and on the other hand its results are used as evidence for the regulation risk assessment performed by RAGRE. It can be considered as the equivalent of software security testing and/or penetration testing acting as an Adversarial Attack Red Team on a targeted AI model and its training/testing/verification dataset. Of course, the STR Testing/Assessment Engine can operate as an autonomous component since it is utilized in various stages of the CoEvolution security assurance flow. The engine considers all the CoEvolution Threat Models and assesses an AI model and its data during design, training and deployment phase applying black box, white box and grey box attacks but also assesses the data bias, the anonymity/privacy of training data as well as possible adversarial poisoning on the training/testing/verification datasets. In that sense, the engine can detect threat against both utility-based adversarial attacks and privacy-based adversarial attacks. More specifically, using as a starting point well-provisioned Adversarial attack libraries like SecML^{xxxii} (created and supported by UNICA), the Adversarial Robustness Toolbox (ART)^{xxxiii} as well as the new research advances on the Adversarial Attack domain we will provide a comprehensive assessment suite that methodically assesses an AI model against most realistic Adversarial Attacks like whitebox and black box Evasion attacks, Poisoning attacks, Model Extraction attacks and Model Inference attacks.

To further consider adversarial attacks on the collaborative learning and fully supply chain paradigm, the security assessment engine will also consider Ensemble Learning attacks that can allow the collaboration of multiple weak adversaries to launch a highly dangerous attack^{xxxiv} on an AI model.

Eventually, in CoEvolution we will design an all-inclusive adversarial data generation mechanism using the latest techniques for creating perturbed data from existing datasets (adopting Generative Learning techniques and focusing on attacks like Adversarial sample Generation and Model Inversion attacks). The goal will be to elevate such mechanism into an AI model fuzzer that will be able to probe black-box AI models (including third party AI models as part of an AI supply chain) and detect evidence of known vulnerabilities to specific Adversarial attacks but provide also indication of unknown “zero day” attacks. The fuzzer will also act as a security evaluation mechanism for the CoEvolution defenses and more specifically the various introduced defence flows in an AI system. In addition, in CoEvolution privacy related attacks are thoroughly assessed in the Security Assessment Engine. While FL and P2P learning solutions by design are supposed to be privacy preserving, in practice by transferring the client model structures or the model parameters, using membership inference techniques the training datasets can be recovered. Hence, in CoEvolution we quantify privacy risks w.r.t. confidentiality of the training data in ML models by information leakage and we use privacy attacks such as membership inference to identify privacy parameters that balance privacy and utility in AI Models. We plan to examine privacy attacks against state-of-the-art domain specific models for classification and prediction (e.g., text or image classification), as well as generative models (e.g., Variational AutoEncoders).

Furthermore, we also consider adversarial attacks that rely on information leakage from the underlining AI model computation infrastructure (e.g CPU, GPU or FPGAs). Such attacks, typically considered blackbox attacks, exploit the side channel information during computation and/or memory usage of the underline computation infrastructure to infer the model structure and its parameters (model extraction type of attacks). To this end we will utilize the ISI partner’s side channel analysis leakage assessment laboratory and adapt it to perform side channel based adversarial attacks assessment. This includes the assessment of single AI models but also the lab will support FL cases where FL aggregation servers are collecting client models on cloud infrastructures especially when such FL servers are

deployed on multi-tenant FPGAs.

Regarding Data assessment in the STR Assessment Engine we will develop methodologies and metrics in order to assess the "safety" of datasets, i.e. whether they have been poisoned or not or they have bias or not. To this end, it will investigate how sanitization strategies employed by a dataset creator can be validated and how their effectiveness can be quantified. Secondly, in CoEvolution we will develop tools which access the sensitivity of models trained with certain datasets with respect to adversarial samples. Since the focus of this task is dataset-centric, such tools will investigate multiple well-known models and measure their sensitivity to adversarial examples when trained with a fixed input dataset. Depending on the dataset and model combination, this task will also investigate whether classic metrics such as the average distance of adversarial samples or new metrics are necessary.

As a final outcome of the STR assessment Engine operations, CoEvolution will provide a series of defined Adversarial Attack metrics that will indicate the exposure of an AI Model to various Adversarial Attack categories. For example, regarding poisoning attack, the CoEvolution Security Assessment engine after assessment will provide percentage values for Poisoned Detection Rate, Poisoned Repair Rate, Data Clean Detection rate, Data Clean repair rate etc. The provided values for all the determined metrics generated by the Engine, will be included in the STR Assessment Engine report. This report along with the risk assessment report will act as the starting point of the STR enhancement procedure that using adversarial attack defences across the whole AI model lifecycle will aim at optimizing the metrics values based on human determined KPIs. Eventually, a final security assessment report will be provided as indicated in Figure 2 security assurance flow where the optimized security assessment metrics are documented. These final metric values will be included in the AI MBOM that accompanies the final, robust AI model and will act as proof of its adversarial attack resistance.

1.2.2.4 Adversarial Attack Defences for STR AI model/Supply chain Lifecycle

As described in the previous subsections, the STR Enhancement consists of three phase categories of techniques, libraries, tools and services to achieve STR defenses following the AI model lifecycle regardless of the Model's adopted paradigm in the AI supply chain. In CoEvolution we propose an innovative approach on how to handle Adversarial Attack defenses which is built around the concept of defense flows and the popular security property of defense-in-depth.

A defense-in-depth **CoEvolution defense flow** consists of series of defenses based on some core AI technology that complement each other in each MLDevOps (AI model lifecycle) phase and are extended as the AI model progresses from one phase to another thus complementing each other. Each defense flow starts from the design phase where a core extendable adversarial attack defense technology is chosen and is combined with complementary defenses to protect against a broader range of attacks compared to the initial technology capabilities. As an AI model designed with the design phase part of a defense flow progresses into training phase (including testing and validation) the provided defenses of the design flow are enhanced (in compliance with the same AI core technology) with training related defenses. The resulting trained AI model now has defenses from design and training phase belonging to the same design flow that harmonically complement each other since they rely on the same AI core technology. However, due to this complementarity, if one of the defenses fails due to some successful adversarial attack then another defense will be present to withstand the attack and keep the AI model robust. The defense flow is enhanced and extended one more time as the AI model progresses into the deployment phase leading to defense flow gadgets associated with the trained AI model thus facilitating the full design flow capacity when an AI model is deployed/operational thus also detect and report adversarial attacks on the operational model. As expected, the deployment defenses added to the design flow increase further (following the defense-in-depth principle) the STR of the associated AI model. The overall design flow constitutes a well-structured manner of organizing and harmonizing different defenses in a consistent manner through the whole AI life cycle. Furthermore, Coevolution design flows can have more than one strands (subflows) as they are extended to various AI architecture paradigms (eg. P2P or FL or interconnected AI agents' cases).

As an example of a CoEvolution design flow, we can consider a flow adopting the Ensemble Learning AI technology that is structured during design phase as an Ensemble of various different base models that collaborate under a voting system. At training/testing/verification phase we extend this Ensemble by added defensive distillation techniques where some base models of the Ensemble are trained as teachers and other as students. Finally, at deployment phase we introduce multithread, parallel processing execution or parallel computation on an edge device FPGA fabric that can further increase robustness at the edge. Note that to manifest a CoEvolution Defence flow each of the three AI lifecycle phase has its unique set of challenges and difficulties that need to be addressed. Eventually each phase contributes to a Defence from in its one way. For the Design Phase, CoEvolution will provide defence flow guidelines for enhancing the AI design, and offer AI models code and defence flow libraries as open-source code that users can incorporate into their model. On the Train and Data Management phase, CoEvolution will provide defence training strategy libraries complementing a defence flow and code for data protection from adversarial attacks, for example by enhancing libraries such as SecML for testing against evasion and poisoning attacks. Finally in the deployment phase various secure gadgets, will be incorporated with the final model in order to provide the AI equivalent of code

instrumentation, runtime mitigation against adversarial attacks, reporting of incidents to the runtime monitoring engine as well as provide update mechanisms to dynamically change deployed model and gadgets to newer versions. The STR defence engine outcomes can be seen in the associated table.

1.2.2.4.1 Design Phase Defences

Overall Approach: Following the Security-by-Design, Explainability-by-Design and Privacy-by-Design Principles at the design phase of an AI model, in CoEvolution we will introduce innovative techniques into the design process of an AI model that will make it resistant to black-box/white-box or grey box adversarial attacks (following the Threat models described in previous subsections). The adversarial attack countermeasures at this phase aim to prepare the ground for the research work to be done in the other phases of the AI model lifecycle and introduce the building block for adversarial attack resistance at the training and deployment phases. The solutions provided at this stage since we are considering also collaborative learning and AI interconnected, full AI supply chain agents, are also focused on structuring the proper interconnections between AI models that can guarantee security and privacy.

Indicative Defence Technologies: Current and novel defense technologies will be assessed and researched during the course of the CoEvolution project, such as the introduction of Homomorphic Encryption (HE) or Secure Multiparty Computation (SMPC) schemes when applicable, to prevent privacy leakage that may allow an adversarial attack exploiting communication link between client and trusted server party (also another peer or another agent in AI supply chain), a reconstruction attack using the paired input and output data, and a membership inference attack by malicious user (an external or internal/client entity). Beyond that, we are also considering at this phase countermeasures to prevent model extraction by side channel attacks (as a black box or grey box attack) both in hardware accelerated or software pre-trained models that an attacker wants to reconstruct (discover the internal structure and possibly the used parameters). Such countermeasures include the use of hiding and/or masking techniques^{xxxv} within the structure of the AI model (eg. in a convolution layer or fully connected layer of a CNN) that effectively prevent an attacker that observes the side channel leakage (eg. power consumption or time related) to identify the operations that are performed and infer the model structure.

Furthermore, in order to provide performance efficient yet robust and secure AI model, in CoEvolution we adopt deep unrolling methods, aiming at transforming well-justified optimization solvers into meaningful and interpretable deep learning architectures that can be robust and explainable to their designers. Deep-unrolling (DU) networks can be many times smaller (thus drastically reducing the amount of the required training data) - while achieving same or higher performance - than their domain-agnostic counterparts, for the task at hand. In CoEvolution we will combine principled mathematical models with data-driven techniques and incorporate domain knowledge leveraged via DU mathematical structures designed for the specific problems related to energy efficiency, performance, and security issues in connected edge AI systems.

Regarding FL based adversarial attacks like model poisoning attacks (assuming a malicious client in the FL scheme), in CoEvolution we acknowledge that traditional FL approaches employing well-known aggregation algorithms like FedAvg fail to capture (inherent or anomalous) heterogeneities among the local datasets, which is especially true in scenarios involving time and space varying conditions (notably, concerning dynamic groups of smart nodes), and in cases involving malicious agents. Also, by design FL schemes rely on a single parameter server that considerably reduces the robustness and reliability of the learning tasks that build on client-server flows. CoEvolution will capitalize on advances regarding privacy-preserving federated learning on non-IID datasets taking into account the data heterogeneity among the participants, based on approaches such as personalized FL (using local fine-tuning and client clustering approaches), and weighted averaging rules, for enhanced system performance and added robustness against adversarial attacks. Furthermore, HE or SMPC schemes will be considered for secure transmission of local and global models (privacy protected from third parties) and secure aggregation without access to the models (privacy protected from the central server). Where HE/SMPC is not possible, alternative, more traditional approaches, will be applied on the FL server by introducing correlation/similarity evaluation on the collected client models acting as an anomaly detection mechanism so that it will be capable of rejecting abnormal/malicious, poisoned, models.

Regarding the distributed learning paradigm solutions (e.g. peer-to-peer learning approaches) using the technology infrastructure provided by TID we provide a solution that does not rely either on differential privacy or on external systems (e.g., blockchain or interplanetary file system (IPFS)). In particular, said solution will consist of a privacy preserving, fully-distributed and infrastructure-less peer-to-peer learning system that (1) enables users to participate in an asynchronous decentralized collaborative learning scheme, benefiting from the wealth of user data that are distributed across the participating devices, while (2) maintaining accurate personalized models. Specifically, the proposed system will utilize partial Homomorphic Encryption (HE) during gradient averaging to preserve both the confidentiality and utility of the shared gradients, while at the same time eliminating information leakage attacks.

1.2.2.4.2 Training and Data Management Phase Defences

Overall Approach: Following the Design Phase, the CoEvolution Training and Data Management Phase defence strategy provides a comprehensive series of complementing defences adopting a defence in depth approach where multiple defences are established into an AI model so that the overall robustness remains high even if some defence

fails. At this phase, the provided defences are related to the training, the testing and the validation process of an AI model but also on the data collection, cleaning and data management to be used as for such training/testing. As in other phases, we consider AI models as white boxes if they are provided after a design phase or as black box models if they are given from a third party and retraining is needed (or transfer learning is supported). Also, we consider both utility-based attacks but also privacy-based attacks.

Model Training/Testing/Validation Indicative Defence Technologies: In the Model training phase as a direct extension of the Design Phase Defences we adopt ensemble methods in the context of adversarial defence, highlighting the significance of encouraging diversity among base models that are ensembled together in order to enhance the overall (ensembled) prediction/inference. In this setup, using the latest development in Ensemble learning we provide an advanced Ensemble Adversarial Attack defence strategy^{xxxvi} that rely on the training of multiple base classifiers to defend against adversarial attacks in a cooperative manner that eventually gets deployed in a parallel processing setup within a computing infrastructure (further discussed in the deployment phase defences). We consider the Ensemble Learning also as a proper defense on a Collaborative training paradigm where Ensemble Federated Learning approaches^{xxxvii,xxxviii} can be considered as a collaboration within the collaboration i.e by training/retraining/updating at each client/peer instead of single model, multiple (two or more) of base models and chose the best of them (or vote for the best of them) to participate to the collaborative learning scheme (e.g to be sent for aggregation at the server). In CoEvolution, we will also consider mechanisms of enhancing Ensemble based Adversarial Attack defences by introducing in them defensive distillation^{xxxix} mechanisms that will create teacher/student networks within an Ensemble to increase the robustness of the voting mechanism. In addition to this enhancement, we are also training of the ensemble models different introduced obfuscation strategies. Combining predictions from these models can increase robustness against adversarial attacks, as the diversity in obfuscation methods makes it more difficult for attackers to craft universal adversarial examples.

Since Ensemble Learning methods may introduce new vulnerabilities, the above-described approach will be supported by more traditional Adversarial Attack Defences. Using as a starting point well-provisioned Adversarial attack mitigation library like SecML^{xl} (created and supported by UNICA) and the Adversarial Robustness Toolbox (ART)^{xli} we will introduce defences against Utility-based attacks and Privacy-based attacks on all AI Model paradigms (single, collaborative learning, Full AI supply chain) based on the latest research developments on Adversarial and/or Robust Training, Regularization, Model Pruning and Perturbing Gradients during the training process^{xlii}. Especially, for the Federated Learning paradigm, the above defences will be applied both to the client and the server and will complement the techniques that have been introduced to the server at the design phase like robust client selection techniques and malicious client detection and removal techniques. Regarding especially privacy-based attacks on Collaborative Learning and Full Supply Chain paradigms, defences are considered primarily on the design phase of such paradigms (eg. HE, SMPC).

It should be noted that this phase is linked with the defences that have been determined in the Design Phase for a given AI Model but also may introduce defences that are extended in the deployment (operational) phase. For example, some run-time detection mechanisms that are deployed as “gadgets” in the deployment phase may have preparatory steps during AI Model training or testing^{xliii}. These activities may include training additional “gadget” components that during deployment will be attached to a run time monitoring gadget.

Data Management Indicative Defence Technologies: To prevent data-related adversarial attacks, in CoEvolution techniques like differential privacy, SMPC or HE and restricted data access that counteract Membership Inference attacks are going to be provided thereby safeguarding the privacy of sensitive data in the training dataset. Preprocessing and anomaly detection methods will be used to protect against Data Poisoning attacks (compromise data integrity), thereby safeguarding the performance and integrity of the AI models from Attacks with backdoors that seek to introduce hidden vulnerabilities into AI systems.

The CoEvolution solutions strengthens input validation and detection systems to thwart evasion attacks during the testing stage, such as those that involve tampering with image data. Security measures are put in place to check input data for anomalies and guarantee its integrity. Regardless of the data type, additional measures are used to identify and reject suspicious or malicious inputs, such as anomaly detection algorithms and input validation procedures. The testing environment's security and dependability are maintained.

1.2.2.4.3 Deployment (Operational) Phase

Overall Approach: The deployment (or operation) phase of an AI Model constitutes the point in the AI model lifecycle post-training up to the point where the model is introduced into an actual system to provide inference/regression for a given problem. At this phase, the various constructed CoEvolution defence flows are completed by adding to them deployment gadgets that can function during the model's operation. Our goal in CoEvolution at this phase is to enhance an AI model with gadgets that have three possible capabilities, a) to be able to detect an inference based adversarial attack on a Model and report it as an incident (security self-awareness), b) to be able to update its defenses when new attacks are discovered by the AI security research community, c) to be able to run in an execution environment that can be considered secure and efficient thus minimizing risks associated with

tainted open-source models and datasets.

In this phase we also consider defences on third-party pre-trained AI models by performing post-training operations (when transfer or continuous learning is supported) or directly introducing at their input or output the CoEvolution add-on gadgets. These approaches help to determine possible poisoned AI models that have not been identified as such during the CoEvolution security assessment following the defence-in-depth CoEvolution strategy. Furthermore, the CoEvolution approach of post-training or adding “gadgets” before an AI Model input or after its output can be applied also on AI models that are secured using CoEvolution assurance flow during design or training phase (following a specific defence flow). While such models are robust enough to withstand inference attacks (like poisoning, evasion or model extraction attacks) the CoEvolution gadgets will be activated to inform the Security Runtime Monitoring Engine that an attack has been attempted on an AI model. This highlights another feature of the CoEvolution gadgets which is their ability to collect information on an attack and report them to the Security Runtime Monitoring Engine thus providing a simplified incident reporting mechanism.

Indicative gadget approaches: Based on these requirements, in CoEvolution, before releasing a trained AI model we enhance it with AI “gadgets” that are AI add-ons capable of increasing further the robustness of the model (post-training) as part of a Coevolution defence flow and/or allow the security self-awareness of an AI Model so that it can detect adversarial attacks. For example we will include Defense GANs and generative learning techniques that introduce synthetic data along with actual data for inference so as to disturb the optimization process of adversarial attacks, making it more challenging for attackers to craft effective perturbations.

In addition, obfuscation at deployment will also be considered in CoEvolution (as part of an obfuscation defence flow). This will act as an extension to the obfuscation defence flow mechanisms introduced in the design and training phases and rely on dynamic obfuscation. More specifically, we are considering model parameter obfuscation or dynamic obfuscation variability in the obfuscation strategies during deployment is introduced in a relevant gadget. For example, we can switch between different obfuscation techniques or parameters periodically to make it challenging for attackers to adapt their methods.

Also, building upon deep unrolling defense flows introduced in the Design Phase, during deployment, an add-on will be introduced to feed a trained AI Model with an enhanced input, exhibiting smaller perturbations than expected by an attack mechanism. Such a “gadget” can be augmented with poisoning attack detection capabilities so that it can determine possible attempts for misclassification. This is accomplished by monitoring differences between raw input features and DU-based features provided by the envisioned add-on. The overall technique can also be used in an FL setup with proper configuration of the FL aggregation server.

Indicative deployment support technologies beyond gadgets: In Coevolution we acknowledge that an AI model is deployed in a real-world system that has specific technology characteristics and that it possibly is participating in an AI supply chain. Thus, the deployment process must take into account a) the specifications of the deployment system (e.g edge device, cloud server, IoT devices etc) and the system’s constraints b) the execution environment exploits that can be used on such system in order to mount an adversarial attack.

Based on the considered CoEvolution Threat models indicative technologies that we will introduce to protect against the above issues are the following:

- On an edge or end device system, an AI model can be easily probed to collect side channel information (power consumption and electromagnetic emission or timing). Thus in CoEvolution, complementing the relevant design phase defenses, we will introduce in the AI model computation flow (in parallel to the AI model execution) side channel attack countermeasures that will make the side channel information useless for an attacker.
- To provide a secure environment for AI model deployment, when supported by the device, we will deploy AI models and their associated gadgets on trusted execution environments (TEE) like ARM trust zone or Intel SGX (secure enclaves). This will provide security and trust guarantees in the execution flow and also prevent privacy-related attacks (complementing the design and training phase already realized techniques).
- Regarding Ensemble Learning Defences, as those are described in the design and training phases defences subsection, in CoEvolution we will further enhance their accuracy by implementing the various base models that produce inference for a voting system in a parallel processing fashion.
- In the Collaborative learning paradigm (as well as in the interconnected AI agents full supply chain) in CoEvolution we consider cybersecurity attacks on the communication flow between peers, client-server or agents. As described in the design phase defences subsection we will introduce at design time HE or SMPC schemes when applicable by the use-case requirements.

Provisions beyond Deployment/Operation: Typically, the AI lifecycle includes two more subcategories in the Operation phase of an AI model. These two subphases are the Operational Update Phase and the Decommission phase. Given the fact that AI Model security may vary as new attacks are discovered and eventually new defences are found to match them, in CoEvolution, at the deployment phase we provide a continuous defences update mechanism. This mechanism is structured around an UPDATE defence flow constructed through the AI lifecycle but primarily completed by using dedicated update gadgets along with the AI model deployment that allow the update

of the AI Model defences through model retraining in order to keep the model's robustness level high. This mechanism is managed by the Security Runtime Monitoring Engine that is responsible for deploying the updates to the AI model. It should be noted that the way this defence flow is designed and implemented may significantly differ according to the used AI architecture paradigm (FL, P2P or interconnected AI agents in a supply chain). The Decommission phase where a model has to be removed from a given AI application in compliance with the proposed EU AI act for high-risk AI, in CoEvolution we design and implement a DECOMMISSION defence flow through the AI life cycle and deploy along with the final robust AI models relevant gadgets that act as decommission proxies to the Security Runtime Monitoring Engine. This engine is responsible for the decommission of the AI model.

1.2.2.5 CoEvolution Trusted Data Generation Simulator

The CoEvolution simulator, based on AVISENSE simulation platform, leverages cutting-edge technologies to create virtual environments where AV software can be tested and validated. It uses physics-based simulations, real-time rendering, and AI-driven scenarios to replicate real-world driving conditions taking into account also realistic network constraints that affect the AVISENSE Collaborative Perception Solutions. While many mature and reliable simulation engines exist for different system components, such as the network or the sensing and physics subsystems, the integration of these simulators into a single framework and workflow is not always a straight-forward task.

Regarding decentralized Peer-to-Peer Learning (P2PL), CoEvolution will take advantage of TID's P2P engine that can deliver realistic simulations of P2PL networks that enables end-to-end communication between peers of the network, either in close physical proximity through proximity sensors or globally via supernodes, rendezvous servers, or a decentralized peer discovery protocol. The P2PL simulator is designed to recreate highly private setups, mimicking cross-device communications over low-bandwidth BLE and WiFi direct, even in the absence of Internet access, to evaluate the performance of P2PL under different network settings, threat models, and ML scenarios, and compare against any compatible baseline models. The integrated simulation engine will be used to study the gains introduced by context-aware solutions against different types of adversarial attacks, and how the context-aware decision support will enable policy optimisation about protocol adaptations and other operational parameters.

1.2.2.6 CoEvolution Biological Inspired Context Awareness Approach

Just as biological receptor cells are adaptable, sensors and probes collecting information in digital networks should be capable of configuration and adaptability at operation time. Thus, AI models operation will remain responsive and resilient in the face of evolving threats, much like the flexibility inherent in biological systems when the environment fluctuates. This precision is vital in both contexts to ensure that the collected information can be effectively *interpreted* and *acted* upon. In the dynamic environments of collaborative AI paradigm (e.g., in P2P and FL) and full AI Supply Chain (Interconnected AI agents), any context-aware AI system would require (i) internalising context at different levels of abstraction, (ii) having basic "sensing" or probing capabilities to derive context information that may be subject to uncertain availability, different actualization periods, and time-limited validity, (iii) interpreting, learning, and classifying contextual information by some mechanism, (iv) reasoning, i.e. deciding what adjustments to the system's behaviour have to be applied according to contextual information. To this end, in CoEvolution we consider the following dimensions for optimal context usage:

Range: In CoEvolution each *context vector* will be constructed on-device using a subset of sensory input (Figure 4, step 1), thus adopting a "limited field of vision" that will (1) reduce the size of the data processed per edge device, (2) make efficient use of the available bandwidth and edge resources (e.g., battery), and (3) reduce latency due to communicating information from a single device across the network to a centralised computing system or cloud server. Specifically, each signal introduced in the context vector (Figure 3, step 2) has an expected adversarial training utility (i.e. reflects the real-time conditions and threats of the operating environment), which will be established in an on-device training fashion by learning policies (Figure 3, step 3) that optimise privacy and security objectives.

Adaptation: The context awareness solution follows a dynamic adaptation strategy, employing genetic algorithms to continuously evolve the context vectors (Figure 3, step 4). This process involves generating a diverse population of context vectors, each undergoing operations such as crossover, mutation, and the creation of offspring. During the crossover and mutation phases that take place at the cloud (in a Central Context Awareness

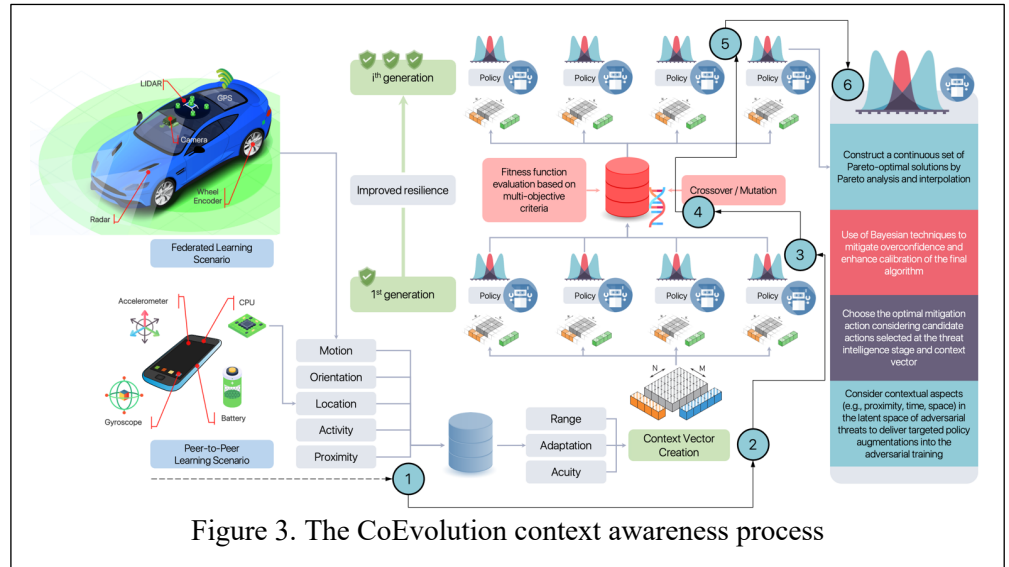


Figure 3. The CoEvolution context awareness process

Engine), the algorithm selects and combines features from different context vectors, akin to a feature selection process. A fitness function, tied to the system's resilience and performance metrics (also based on the learned policies in step 3), is utilised to evaluate and select the fittest vectors for propagation to subsequent generations, which are then redistributed to the edge devices, fostering a continuous cycle of improvement and adaptation. The resulting offspring from these genetic operations enter a new evaluation cycle, which is closely tied to the learned policies and how they impact system's resilience (Figure 3, step 5). Through this evolutionary approach, **CoEvolution** will maintain an optimal level of context-awareness, resulting in a more resilient and efficient P2P and FL network environment

Acuity: In acuity of the sensory information embedded in a context vector will be derived implicitly by means of policy learning (offline or using the CoEvolution Simulator) by the **CoEvolution Context-Aware Decision Support (CDS)** component (described in details below).

At the core of the context awareness module, infused as a gadget at the deployment phase of CoEvolution STR defences flow in each AI model, lies the **Context-Aware Decision Support (CDS)** agent that operationalises a wide array of information from the environment (Figure 3, steps 1-3), effectively sensing and probing, to extract actionable insights and choose the optimal mitigation action as those are suggested by RAGRE (Figure 4, step 6). This process is crucial for identifying emerging threats, adapting to changes in the network, user behaviour, and potential security challenges head-on. The CDS module will use the generated by the Central Context Awareness Engine context vectors to learn the optimal policy about protocol adaptations and other operational parameters. The CDS will utilise an effective representation for obtaining the best performance trade-offs for multi-objective decision support by means of a Pareto set, which will be composed from different policy families. The CDS objective is twofold: (1) to maximise the privacy of data exchanges and (2) to fortify the security protocols in place. CDS dynamically adjusts its policy based on the context, selecting optimal actions such as: (1) Adapting communication protocols to mitigate the risk of data breaches or eavesdropping; (2) Strategically connecting with trustworthy Peers, thereby reinforcing the reliability of the learning network; and (3) Adapting the frequency and granularity of model updates shared across the network to balance privacy with collaborative learning efficacy.

1.2.2.7 CoEvolution Security Runtime Monitoring

The CoEvolution AI Security Runtime Monitoring Engine (SRME) has a dual role in the CoEvolution hub. Firstly, SRME continuously monitors the AI Model security status by collecting events (e.g adversarial attack incidents) from the AI Model deployed gadgets and use that information to associate them with known adversarial attacks categories. Eventually, the above reporting information can be consolidated as an incident report that can be linked with the AI Model through its AI MBOM. Secondly, the Security Runtime Monitoring Engine is managing the AI MBOMs that are generated through the security assurance flow. In practice, the SRME is acting as a AI MBOM management trusted authority that is entitled to evaluate the validity of an AI MBOM (i.e check that all AI MBOM fields are valid) and to assess the AI MBOM integrity and authenticity. Failure to validate will result in disqualifying the AI model from use.

Regarding the first functionality of the SRME denoted as **SRME awareness mechanism**, typical secure communication flows are used to transfer the collected evidence on inference adversarial attacks as those are detected by the deployment phase AI model gadgets. Those gadgets represent the end point of the defense flows as those have been realized through the whole AI model lifecycle. It should be mentioned that the provided adversarial attack evidence can be interrelated with each other since AI models can be considered as part of a supply chain especially

in collaborative learning paradigm setup (P2P or FL) or in an interconnected AI agents setup. In these cases, the evidence of an adversarial attack incident may constitute a cascading attack that affects the full supply chain and it is reported to a human user through an CoEvolution Dashboard. Eventually, the SRME acts as the ultimate endpoint of a defense flow where incidents affecting the flow are collected (i.e attacks that are targeting the defense flow) and reported. Finally, those incident reports can be considered as regulation risk evidence that can lead to an update of the regulation risk status (in compliance with the proposed EU AI act) of a given, attacked, AI model or model supply chain. This evidence after been evaluated and associated with adversarial attacks are forwarded to the CoEvolution risk assessment, guidelines and recommendation engine and may lead to the model's AI MBOM update.

The second functionality of the Security Runtime Monitoring Engine, SRME, is to manage the AI MBOM pivotal component of the CoEvolution solution (**SRME AI MBOM Management Mechanism**). Thus, the SRME is associated with the AI MBOM pool, where all the created AI MBOMs are stored, with the risk assessment, guidelines and recommendations engine as well as the CoEvolution assurance process (responsible for issuing or re-issuing an AI MBOM). The main service of the SRME AI MBOM Management Mechanism is to reply to requests regarding the validity, authenticity, integrity of an AI MBOM. The SRME has all the necessary security mechanisms like digital

signature schemes and authentication mechanisms in order to validate the integrity and authenticity of the AI MBOM, identify the authentic issuer and the information that are included in it. This process is performed by using traditional digital signature schemes associated with PKI certificates by the issuing and verifying authority. Considering that each AI MBOM must be associated uniquely with an AI Model within a supply chain, the SRME AI MBOM Management Mechanism always validates the unique identifier field within the

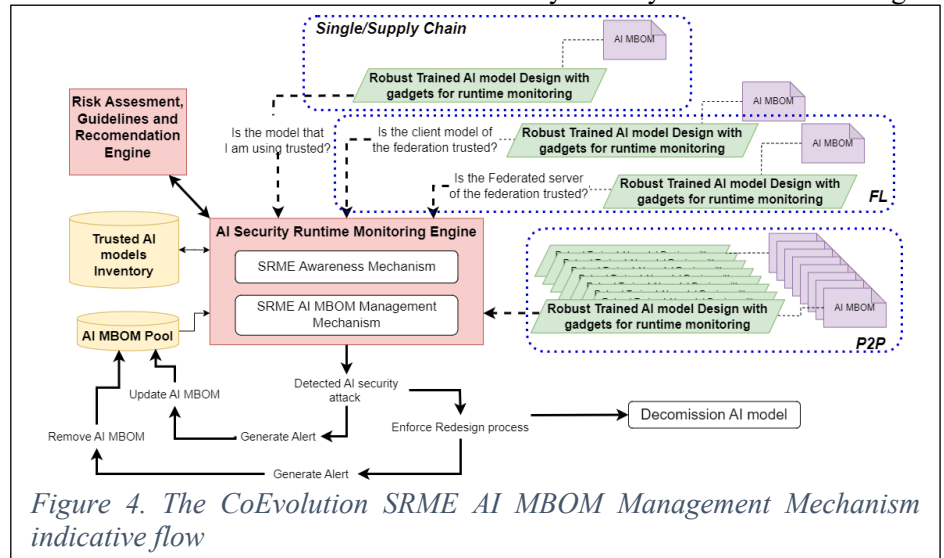


Figure 4. The CoEvolution SRME AI MBOM Management Mechanism indicative flow

AI MBOM with the AI Model generated identifier value using the IDENTIFY defense flow gadget included in the AI Model during deployment. Using the SRME AI MBOM Management Mechanism, an entity within the AI supply chain can retrieve a third party AI model's AI MBOM (including another peer or client/server in a collaborative learning setting) and provide it to the user of this model, it can also validate that a provided by a third party AI MBOM is legitimate thus the AI model associated with it can be used, if the security and regulation requirements set by the model user are matched (based on the AI MBOM provided information).

If inconsistencies are found between the AI MBOM and the associated AI model are found then the AI MBOM is revoked and the model needs to be decommissioned using the DECOMMISSION defense flow gadget on the AI model. The SRME AI MBOM Management Mechanism also has an AI MBOM update mechanism that allows the MBOM to be updated when the associated AI model gets updated. The overall flow of the SRME AI MBOM Management mechanism is presented in Figure 4.

1.2.2.8 AI MBOM Concept and Design

AI MBOM is a critical component of the CoEvolution concept since it constitutes the building block for certifying the trust, robustness, and security of AI systems that have adopted the CoEvolution ecosystem. Introducing into the AI world, the Bills of Material (BOM) concept found in software engineering and Software SecDevOps we can create a new comprehensive structure (the AI MBOM) that captures the essence of an AI model design and development but also is enhanced with all the necessary security, trust, robustness and explainability reasoning that will allow an AI model user to trust this model within an collaborated learning setting and/or as part of an AI model supply chain. The AI MBOM is uniquely associated with an AI model and provides comprehensive information about it, including the original training dataset metadata, poisoning/bias mitigation algorithms and model evaluations. This comprehensive record-keeping ensures that AI models are thoroughly vetted, and their development process is transparent and accountable. AI MBOM's ability to update records with new data enables AI systems to remain up-to-date and adaptable to evolving environments. Additionally, AI MBOM facilitates the involvement of humans in the loop by providing a platform for continuous monitoring, testing, and evaluation. This allows for human intervention when required to maintain control and trust in AI systems.

More specifically the primary objectives of the CoEvolution AI MBOM are:

- **Transparency:** Providing clarity on the tools, hardware, data sources, and methodologies used in the design and

development of AI models.

- **Reproducibility:** Offering enough information for researchers and developers to assess the validity of the models and results and eventually be able to reproduce them with respect to its security and access control capabilities posed by the Model creators.
- **Accountability:** Ensuring creators and users of AI systems are aware of their origins, components, and performance metrics.
- **Assessment-Audit:** Report discovered adversarial attack vulnerabilities, exploits and associated threats under specific threat models along with their associated risk. Also report if they have been countered through CoEvolution
- **Security and Trust:** Provide adequate information on the countermeasures that have been applied through the AI model lifecycle to protect the model from adversarial attacks.
- **Ethical and Responsible AI:** Encouraging the documentation of training data sources, including any synthetic data used, to ensure there's knowledge about potential biases, limitations, or ethical considerations in line with EU guidelines for trustworthy AI
- **Uniqueness:** Provide evidence on the authenticity of the model and associate uniquely the AI model with the associated AI MBOM
- **Risk Assessment:** Provide and associated risk for security, privacy, safety, robustness as well as the regulation related risk in accordance with the proposed EU AI Act
- **Human in the Loop:** Facilitate the involvement of humans in the loop, enabling human intervention when required.

The AI MBOM however, as proposed in CoEvolution is more than a traditional Software BOM adaptation to the AI domain. It supports additional features related to robustness, fairness, security, trustworthiness that are in line with the EU activities on Artificial Intelligence, such as the proposed Artificial Intelligence Act. The AI MBOM through the CoEvolution Risk Assessment, Guidelines and Recommendations reports can include multiple risk levels (in line with the EU AI Act). For these reasons, the AI MBOM structure will have dedicated fields that will support the above-described concepts.

However, the AI MBOM should be also viewed from an AI security and resilience/robustness perspective. It is uniquely associated with an AI Model, it documents the various defense flow that such a model is using, provides a thorough report of the security assessment metrics values (security assessment score) that it has achieved and also in relation to the regulation risks associated with the model, it documents the context under which it can operate. Thus, by design the AI MBOM provides a plethora of information on the model structure, capabilities, training datasets, discovered security/adversarial attack vulnerabilities and their mitigation measures etc that can be used by an AI adversarial attacker or any unauthorized party to perform a broad range of adversarial attacks. This, by definition, counters the concept of closed AI models (that act as black boxes). Hence, the AI MBOM requires a several access/authorization tiers on which entity is entitled to view the Security and privacy related MBOM information.

The tiered system of access rights to an AI MBOM applies mostly to closed-sourced models that will be evaluated by the CoEvolution solution and any open-source models that do not provide their internal information available to the public, but only their final product. The CoEvolution envisions a three-tiered system, with each Tier providing different levels of visibility in the AI MBOM details, respectively to the access and authorization right an entity is entitled to.

The first Tier, Tier 1, is visible to everyone and simply provides a Security Scrutiny Guarantee, that a specific model has been thoroughly vetted and tested by the CoEvolution solution. Tier 2 provides all Assessment and Audit Results, as well Risk disclosure without providing any details about the internals of the model within the AI MBOM. The final Tier, Tier 3, users have complete access to the information residing in the AI MBOM. The AI MBOMs produced by the CoEvolution solution are stored in a shared repository that multiple organisations, end users can use. The repository (AI MBOM pool) is managed by the CoEvolution Security Runtime monitoring engine which has the authority to re-sign, update or revoke AI MBOMs following defined governance procedures and policies so that any unauthorised access can be prohibited.

1.3 Pilots and Use cases

1.3.1 Pilot 1 Securing the AI Supply Chain of Manufacturing Infrastructure

1.3.1.1 Problem Statement

In industrial automation, AI components are increasingly important and used to automate for example the management of processes, machine control, or logistics as well as the predictive maintenance of deployed hardware. Often these AI components are brought into the manufacturing environment as part of apps that are executable in an edge computing infrastructure. These apps can be downloaded from marketplaces of edge ecosystems. An app can thereby be provided to the marketplace by third-party vendors. The AI component of the app may have been even developed by another subcontracted party that is supporting the third-party vendor.

Hence, for the end-customer, the operator of the manufacturing plant, it is crucial that security mechanisms are in place to be able to trust the app and its contained AI component. Especially, when apps are part of essential workflow chains (consisting of multiple apps), it is of utmost importance to ensure their security and expected behaviour.

1.3.1.2 Description of the existing infrastructure and use cases

In this project, we focus on connected AI components that can be used to manage the logistics of the manufacturing environment as well as the management of the network and edge infrastructure itself.

As illustrated in the figure above, we assume an AI model comes in (e.g., as part of an edge app) from a third-party vendor or other facility. This is either a pretrained AI that still requires a transfer learning step to be adjusted for the new environment, or it can be used as it is.

To implement the use case, the following components are involved:

Network/Edge Management AI: This AI model supports the management of the network / edge infrastructure, and it can for example be utilized to efficiently manage edge resources and installed apps.

Network switch devices: These devices allow the creation of connected edge networks.

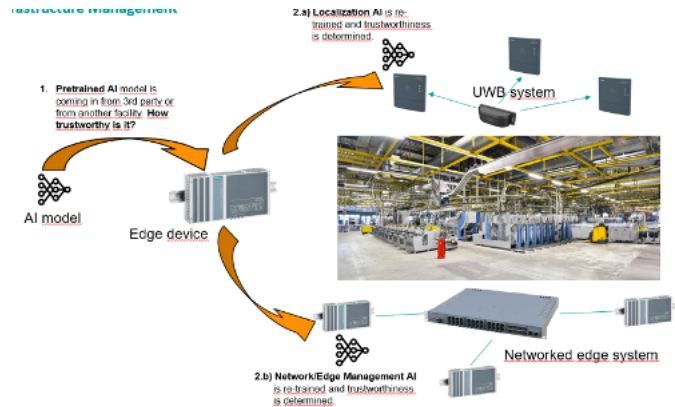
Edge device: This is the device on which an AI model will be deployed.

Localization AI: This AI model supports the logistics of the manufacturing environment, and it can for example be utilized to improve the localization accuracy of tracked assets.

Localization beacon devices: These devices enable wireless localization of tracked assets.

Localization tag devices: These devices are being localized by the beacons and carried by the tracked assets.

In both cases, it is crucial to be able to trust the AI component. Adversarial attacks that could harm the localization accuracy or the efficiency of the edge resource management, could have substantial negative effects on the productivity of the manufacturing plant.



KPIs/Success indicators

1. Delivery of 2 use cases in this pilot (AI-based localization for asset tracking in industrial logistics and AI-based edge/network management), which will be secured through the CO-EVOLUTION framework.
2. Successful conduction of a security training focused on the CO-EVOLUTION framework for the personnel of the organization driving the use cases.
3. At least 80% of the stakeholders involved in the use case express a positive view on: the CO-EVOLUTION framework helps to increase the trustworthiness in AI components (validated through questionnaires).

1.3.2 Pilot 2 Securing the AI Supply Chain in the Autonomous Vehicles Domain

1.3.2.1 Problem Statement

CAVs utilize a multitude of onboard sensors, including camera, radar, LiDAR, ultrasonics, and GPS to establish a comprehensive understanding of their surroundings. However, the employed sensors have considerable inherent limitations (e.g. weathering and/or lighting conditions degrade camera performance, while active sensors are adversely affected by high humidity, occlusions, and interference), which can severely degrade CAVs performance. What is more, in real-world scenarios, the amount of local data is insufficient to derive resilient and robust learning models, due to the extremely complex and dynamic nature of the CAV's environment. Cooperative perception aims to overcome these adversities by sharing knowledge among a group of interacting agents in order to b) robustify the CAVs' environmental modeling capabilities; b) extend CAV's local sensing range; and c) enhance the system's resilience to sensor failures/attacks.

1.3.2.2 Issues/Problems with the existing approaches

Recent studies^{xlv} show that how that even state-of-the-art Deep Neural Networks (DNNs), which constitute instrumental components for AI-based perception CAVs, are vulnerable to adversarial attacks, even in a black-box setting. In addition, several works^{xlv,xlvi} demonstrate that potential data-poisoning attacks such as LiDAR jamming,



or GPS spoofing, can cause vehicle's control unit to behave erroneously. Recent grave examples^{xlvi, xlviii} of malfunctioning autonomous vehicles serve as a dire reminder of the level of accuracy, robustness, and resilience required for the reliable operation of autonomous vehicles in order to eliminate safety risks for everyone involved.

1.3.2.3 Description of the existing infrastructure and use cases

The use case will take place at PASEU's premises in Langen, Germany and will validate the development in a variety of **valet parking setup configurations**, in both urban and suburban environments, under a) Semi-automated, and b) Fully automated parking assistance scenarios. In the first case, the driver remains in the car and receives suggestions for free parking spots and optimal parking maneuvers, while in the latter the driver leaves the vehicle at the beginning of a parking area so that both area exploration for free parking spots and the required maneuvers are performed autonomously. To this end, PASEU brings its cutting-edge CAV testing infrastructure, offering a) Highly precise maps and LiDAR, surround view omnidirectional sensors operating at the visual spectral band additionally stereo camera on the dash board from far range sensing and suspension control; b) a Hybrid-EV test vehicle (Toyota-Prius) equipped with full sensing and data acquisition setup, where the software for the trajectory planner and control is fully developed and control c) a dedicated area for AD functions testing consisting of various configurations of road structures and traffic conditions ranging from urban areas, to intra-urban as well as motorway sections; d) mobility infrastructure supporting AD functionality requirements. AviSense brings its comprehensive suite of perception-oriented solutions comprising of a) the data enhancement Kit for enhancing the quality of low-cost visual and LiDAR sensors; b) the Hyper-Sensing Kit for delivering trusted multi-agent perception localisation and mapping; c) the comprehensive CAV Simulation Platform, specifically crafted for developing, validating, and testing Autonomous Driving software; d) highly innovative XR system of xAI content, including IoT devices with an integrated V2X stack. **Evaluation Scenarios:** This use case will test CoEvolution's cyber-security engine in multi-agent scenarios involving CAVs that cooperate to establish autonomous navigation and co-operative multi-modal four dimensional scene awareness in the vicinity of the vehicle. To this end, the use case will adopt early (sensory inputs) and late (AI-leveraged knowledge) fusion approaches to introduce a holistic unified framework for co-operative awareness and Multi-Agent-Path Planning (MAPP), jointly interconnected in an iterative loop. For the co-operative awareness phase, CoEvolution will integrate data derived by both mobile and static Interacting Traffic Agents (ITAs), where static ITAs can consist of both sensors lying on the infrastructure as well as landmarks and active sensors, acting as landmarks. In alignment with CoEvolution's principles and objectives through this use case, we will develop a trustworthy, robust and context-aware cyber resilience approach, where the concept understanding will be derived by cooperative scene understanding and path planning. Our key insight is that both sensing and planning problems are inherently variants of accurate geo-localization, which is vulnerable to cyber-attacks on multiple surfaces (GPS-spoofing, camera, vehicles ethernet). Therefore, we consider the joint sensing-planning solution to eliminate the effects of added noise in the traditional two step approach and deploy a sophisticated joint optimization framework, resilient to cyber-attacks. The idea is to compute the complete continuous-time trajectory from start to goal in a synchronous manner, such that given the current time-step, the solutions to the estimation problem (history of the scene) leverages context awareness and the planning problem (prediction of scene evolution and trajectory) automatically fall out. Additionally, performing this joint optimization allows information to flow between perception and planning resulting in mutual benefits. This is approach highly innovative as the CAV must contend with a potentially high-degree-of-freedom (DOF) trajectory space, as well as to model the uncertainty due to potential cyber-attacks, and the stochastic effect of executing actions by the controller. The pilot will offer a comprehensive testbed for validating and demonstrating the CoEvolution ecosystem under a wide variety of threat models encompassing all AI life-cycle stages. More specifically: a) the **data acquisition** stage will involve generating simulated and real multi-sensory training data covering a wide range of attack scenarios, and will feed all subsequent cyber-resilience tests; b) the **design** stage will involve designing wide range of deep architectures for image/point cloud quality enhancement, multi-modal multi-agent object detection, feature extraction, and others, utilizing both third-party pretrained networks, and initial designs utilizing deep-unrolling approaches; c) the **training** stage will include isolated/standalone model training using only local data, and HE-enabled collaborative training under the FL paradigm; and d) the **operation** stage will involve monitoring the deployed models to verify and mitigate potential adversarial attacks utilizing CoEvolution's add-on gadgets based on the multi-modal collaborative AI paradigm, exploiting the potential and limitations of joint co-operative scene understanding and MAPP in mitigating cyber-attacks.

KPIs/Success indicators

1. KPI-UC2.1: Efficiency in mitigating the attack in terms of the CAV's geo-localization
2. KPI-UC2.2: Deviation of the originally planned trajectory planning towards the one after the mitigation of the attack;
3. KPI-UC2.3: Curvature of the trajectory planning before and after the attack;
4. KPI-UC2.4: Energy Overhead in terms of the computational resources allocated to mitigate the attack.
5. KPI-UC2.5: Overall energy footprint caused by the attack involving the energy allocated by computational

resources plus by the vehicles motion.

6. KPI-UC2.6 Deterministicability of the mitigation strategy in terms of the CAV planning after the attack

1.3.3 Synergies with relevant European and national Initiatives

Projects	Relation to CoEvolution
HEU-SECURED [ISI]	Utilize the scalable privacy Preserving (HE/SMPC) libraries developed in SECURED in order to created defences against privacy-based adversarial attacks
Avalanche [UNIFI]	Avalanche will be used to develop incremental security defenses via security patches and study attacks in incremental learning (T4.5).
AI-Toolkit [UNIFI, ICCS]	The components and design philosophy of an AI Toolkit can be exploited to implement CoEvolution applications.
HEU-EMERGE [UNIFI]	EMERGE studies distributed learning and continual learning, including efficient and untrained networks, which are relevant for CoEvolution objectives.
H2020-I-BIDAAS [AEGIS, TID]	I-BiDaaS builds upon data sharing and breaking silos in industrial environments; know-how, lessons learned, and best practices from I-BiDaaS can be transferred to CoEvolution.
H2020-CONCORDIA [TID]	CoEvolution will draw knowledge from CONCORDIA, to exploit both its technologies and cybersecurity ecosystem. TID will bring to CoEvolution scalable, privacy-preserving technologies in collaborative learning, among others.
H2020-ACCORDION [TID,ICCS]	TID and ICCS will bring in from ACCORDION various R&D efforts on private and leakage free ML technologies in collaborative learning, e.g., FL, Hierarchical FL, etc.
H2020-SPATIAL [TID]	TID will bring in R&D efforts from SPATIAL on privacy-preserving, trustworthy ML technologies, as well accountability, resilience and privacy metrics and AI algorithms. Also various Algorithmic solutions to ML inference attacks.
H2020-HEIR [IMT]	Use the aggregator and analyzer of vulnerability sources developer in H2020 HEIR in order to assess data quality.
HEU-ELSA [UNICA]	CoEvolution will build on the ELSA project by developing tools for improving robustness of AI models that are continuously updated and deployed.
HEU-Sec4AI4Sec [UNICA]	CoEvolution will build on the Sec4AI4Sec project to improve AI security testing and robustness in a continuous development and operations setting.

1.3.4 Inter-disciplinary approach

CoEvolution scientific disciplines are recognized to be Science, Technology, Engineering and Mathematics (STEM) but with a clear integration with Information & Communication technologies, and it is committed to use the project's outputs to shape the policy strategy and plans of the European Commission. CoEvolution applies a collaborative research framework to interlink the expertise and methods from those different disciplines in pursuit of the CoEvolution objectives. In specific, CoEvolution focuses in the internal and external alignment of goals and ambitions and their conversion to practical opportunities and possibilities based on the extensive experience of the coordinator and the consortium partners. It ensures social interaction and social legitimacy through the inclusion of diverse participants with complementary expertise (Section 3.2.3) and the creation of a level playing field, in which relationships of trust can be built to negotiate and deliberate ideas and perspectives. CoEvolution also focuses on building the necessary skills and competences therefore creates and facilitates engagements that allow social learning, scale up transition and productive interaction in relevant fields in society for producing impact.

1.3.5 Social Sciences and Humanities (SSH)

CoEvolution while is does not directly build on Social Sciences and Humanities (SSH) concepts it will take into account as a consortium the stakeholder engagement, participatory methods, living labs, etc., to figure out the impact and the upgrade potential that the different solutions and disciplines bring in the project. CoEvolution intends to capture and present how these SSH disciplines are integrated in the project in terms of activity type, budget, countries, sectors, experts, impact, based on a carefully-drawn selection of criteria, indicators and parameters.

1.3.6 Sex and Gender Analysis

CoEvolution completely supports the European Commission's Gender Equality Strategy for 2020-2025 and endorses the recommendations and provisions made public to Horizon Europe in the report by the Commission's Expert Group on Gendered Innovations under the coordination of DG R&I's Gender Sector, entitled: "Gendered Innovations 2: How inclusive analysis contributes to research and innovation". In CoEvolution, in each phase of the adopted methodology, the research team (including engineers, data scientists, ML experts, SSH, ethics and legal experts, use-case participants) will be required to consider gender-specific aspects that may arise in the planned work. In CoEvolution methodology, the team will identify any gender aspects that should be considered during the state-of-art analysis (including possible gender-induced biases in the employed datasets) and the definition of the CoEvolution

user requirements. Subsequently, during development of the AI-related techniques, their initial evaluation will take into account and report upon gender aspects that may arise, not only due to possible dataset biases, but also due to human-machine interactions. Finally, during CoEvolution validation in the planned use-cases, gender aspects will be considered in the evaluation procedure both from a technical (e.g., gender-induced bias impact on the observed system performance) and a social (e.g., gender-aware, end-user trust and acceptance) stand-point and best practices will be documented and communicated.

1.3.7 Open Science

The project will actively promote the openness of results to align with the guidelines and recommendations in EU-funded projects, as well as any national policies for open data. Targeted activities are foreseen in WP2 towards open scientific publications (“green” or “gold” open access model), open research data, software code (in open or closed form) and reproducible research (to allow researchers to reproduce the results online using the open datasets and code, but without the need for specific software licenses and/or specialized hardware on their computers). In particular, the scientific publications will be self-archived in Open Access repositories (i.e., the European OpenAIRE platform) to provide easy and early access to the research community. Selected data related to the measurements and research results stemming from these publications will also be openly available through open platforms after investigating potential IP exploitation options. Software code implementing the envisioned algorithms and methods developed in the project will be released either as open or closed source depending on the respective IP exploitation plan. In the open-source form, the source code of the software will be released under public access, while in the closed source form, the compiled version of the code will be available. In both cases, the software will be linked to the associated scientific publication. Finally, to promote and actively support reproducible research, the CoEvolution team will use open research collaboration platforms for publishing the open software code and open dataset(s) to allow researchers to reproduce online the results presented in the associated publication.

1.3.8 Research Data Management

The project involves carrying out data collection and a set of demonstration tests will be carried out to assess the technology and effectiveness of the technical solution in 2 pilots. Any processing of personal data will be conducted in accordance with the provisions of: **a)** the GDPR (Regulation (EU) 2016/679), **b)** the Universal Declaration of Human Rights and the Convention 108 for the Protection of Individuals with Regard to Automatic Processing of Personal Data, and **c)** the national laws applying its provisions, including those governing the acquisition of valid consent. CoEvolution will establish a Data Management Plan (DMP - D1.2 M6) which will detail what data the project will generate (i.e. content, type, format, volume), which standards and methodologies will be used for data collection and management, whether and how it will be exploited and how they will be made findable, accessible, interoperable and reusable (FAIR). Management of research data will be done in accordance with the related soft law instruments governing scientific research, such as the European Code of Conduct for Research Integrity, the Guidelines to rules on Open Access to Scientific Publications & Open Access to Research Data in Horizon Europe, and the Guidelines on Data Management in Horizon Europe. Indicatively, the types of data/research outputs concern measurements of runtime monitoring, security, energy, and performance optimization data in several forms such as numerical, texts, maps, images, tables, etc. and during the project will be deposited in trusted **Open Data repository** repositories as [Zenodo](#), which assigns DOIs ensuring their **findability**. To ensure **reusability of data/research outputs**, the data will be well-documented and will have clear license and provenance information. Indicatively, CoEvolution will use the Creative Commons licenses indicatively, the [Attribution \(CC BY\)](#) license and [Creative Commons Zero \(CC0\)](#) to dedicate works to the public domain.

2 Impact

2.1 Project’s pathways towards impact

The impact maximisation approach for the CoEvolution project relies on the following pillars:

- **Raise awareness:** Promote the CoEvolution project and its benefits through various channels, such as conferences, workshops, publications, and social media, targeting AI developers, researchers, and industry stakeholders (**WP2, T2.1**).
- **Collaborate with stakeholders:** Engage with key stakeholders, including AI developers, researchers, industry partners, and policymakers, to ensure the project's results align with their needs and expectations (**T2.3**).
- **Develop strategic partnerships:** Establish partnerships with relevant organizations, research institutions, and industry players to facilitate the adoption and integration of the CoEvolution project's results into existing AI ecosystems.
- **Standardization and compliance:** Work towards influencing and shaping AI-related standards, guidelines, and regulations to ensure the CoEvolution project's results are aligned with the broader AI community's goals and objectives (**WP1, T1.5**).
- **Education and training:** Provide training and educational resources to help AI developers, researchers, and industry professionals understand and adopt the CoEvolution project's security-by-design, privacy-by-

design, and explainability-by-design principles (WP2, T2.5).

- **Continuous improvement:** Monitor the project's progress and gather feedback from stakeholders to identify areas for improvement and ensure the project's results remain relevant and effective in addressing the evolving needs of the AI community (WP1, T1.2).
- **Commercialization:** Develop a viable and well-structured business plan for the exploitation of the CoEvolution solution, ensuring its sustainability and widespread adoption beyond the project's duration (Section 2.2.3).

2.1.1 Analysis of target stakeholders and expected benefits

CoEvolution targets a diverse set of stakeholders affected by the inclusion of context awareness in machine learning to enhance resiliency and the concept of security-by-design and resilience to adversarial attacks.

- **AI Researchers and Developers:** CoEvolution provides them with tools, techniques, and frameworks to build more secure, robust, and trustworthy AI models and systems to mitigate risks like adversarial attacks, data poisoning, model extraction etc. The broader AI research community can benefit from CoEvolution's initiatives like the AI MBOM data structure and momentum towards a robust and secure AI supply chain ecosystem.
- **Security Professionals and Researchers:** Security-by-design and resilience to adversarial attacks are crucial for security professionals and researchers in the field of network security, as this is explored by CoEvolution. Adversarial attacks pose significant threats in areas such as intrusion detection and malware detection, where machine learning is increasingly being adopted for such tasks.
- **Government and Regulatory Bodies:** Evidence of adversarial robustness evaluation is recommended as a minimum requirement for the acceptance of AI-based security systems. This implies that government and regulatory bodies play a crucial role in setting standards and evaluating the robustness of AI-based security systems to adversarial attacks.
- **Technology Companies:** Technology companies and organizations developing or deploying AI solutions can benefit from CoEvolution by leveraging its comprehensive framework to secure their AI supply chain. This helps them address interoperability, security, and data integrity challenges.
- **Software and Hardware Vendors:** Companies that produce software or hardware for other organizations are directly affected by the need to enhance security in their disruptive technologies.
- **Industry and Business Leaders:** The inclusion of context awareness in machine learning can impact various industries and businesses, especially those reliant on AI technologies. Understanding and leveraging contextual awareness in AI systems like the approach followed in CoEvolution, can lead to more robust and resilient applications across sectors.
- **End Users of AI Systems and General Public:** CoEvolution aims to provide end users (e.g. in healthcare, transportation, finance etc.) with more reliable, safe, and trustworthy AI systems that adhere to ethical standards. The resilience of security systems to adversarial attacks directly impacts end users and the general public.
- **Policy Makers:** Policy makers and regulatory bodies working on AI ethics and governance can utilize CoEvolution's frameworks aligned with EU regulations on trustworthy AI. This aids policy making around ethical and secure AI.

2.1.2 Contribution to the outcomes of the topic and to wider impacts

CoEvolution envisions to significantly contribute to the **HORIZON-CL3-2023-CS-01-03** call's expected outcomes and wider impacts in the following ways:

Expected Outcome #1: Security-by-design concept and resilience to adversarial attacks

The CoEvolution project aims to make a unique contribution to the **security-by-design concept and resilience to adversarial attack**, by enhancing AI models with "gadgets" that increase their robustness and security self-awareness. These gadgets, such as Defense GANs and generative learning techniques, disturb the optimization process of adversarial attacks, making it more challenging for attackers to craft effective perturbations (**Obj2**). Additionally, CoEvolution includes obfuscation at deployment, which introduces variability in obfuscation strategies to make it challenging for attackers to adapt their methods. The approach provides security enhancement and robustification in all steps of the AI model lifecycle, including design, training/testing/validation, and deployment/operation phases (**Obj1**). CoEvolution also infuses AI models with context awareness, allowing them to respond differently based on the environment in which they are deployed and to monitor their security status, detecting and reporting attacks to the cybersecurity runtime monitoring system (**Obj4**). The CoEvolution approach considers various threat models and offers a comprehensive assessment suite to methodically assess AI models against realistic adversarial attacks, including ensemble attacks. CoEvolution prioritizes security throughout the AI model lifecycle, providing guidelines, defence training strategies, and secure gadgets for runtime monitoring, mitigation, reporting, and updates (**Obj6**). This approach also introduces innovative techniques into the design process of AI models to make them resistant to adversarial attacks, including collaborative learning and

interconnected AI supply chain agents. CoEvolution assesses and researches defence technologies, such as Homomorphic Encryption and Secure Multiparty Computation, to prevent privacy leakage and model extraction. [**Scale:** *Cybersecurity Industry, (Cyber-) security Professionals and Researchers, AI developers and Engineers, Government & Regulatory Bodies, Organisations/Industries relying on AI applications* **Significance:** *Addressing of critical concerns related to data privacy, model integrity and system effectiveness, Enhancing security, robustness, and context-awareness of AI models*]

Targets	Adversarial Attack Detection Rate by the AI models enhanced with CoEvolution "gadgets" and security self-awareness > 95% (T6.4)
---------	---

Expected Outcome #2: Inclusion of context awareness in machine learning in order to boost resiliency

CoEvolution contributes to the **inclusion of context awareness in machine learning to boost resiliency** by infusing AI models with context awareness throughout their lifecycle (**Obj4**). The project's approach involves internalizing context at different levels of abstraction, basic sensing or probing capabilities to derive context information subject to uncertain availability, and interpreting, learning, and classifying contextual information. CoEvolution's context-awareness module, infused as a gadget at the deployment phase of the CoEvolution STR enhancement engine in each AI model, collects a wide array of information from the network, effectively sensing and probing to extract actionable insights (**Obj5**). This process is crucial for identifying emerging threats and adapting to changes. CoEvolution's context-awareness mechanism is supported by a unique data structure, the AI MBOM, which provides detailed information about an AI model, including security-related information associated with the use of the CoEvolution solution. The project's long-term goal is to act as an enabler to a concrete ecosystem around security and trust in AI solutions, capturing different aspects of ML/DL research and development throughout their lifecycle [**Scale:** *Organizations, Stakeholders utilizing AI technologies for diverse applications, Significance: Boosting resilience of AI systems through inclusion of context awareness, AI systems adaptability, effective interpretation and acting upon environmental stimuli*]

Targets	Context-awareness effectiveness index calculated based on the successful adaptation of AI models to changing environments, the accuracy of security status monitoring, and the timely detection and reporting of attacks >85% (T5.5, T6.4)
---------	--

Destination Impact #1: Strengthened EU cybersecurity capacities and European Union sovereignty in digital technologies

The CoEvolution suggested framework incorporates several key elements that align with the EU's goals **to strengthen cybersecurity and assert digital sovereignty**. CoEvolution provides robust tools, techniques, and guidelines for hardening AI systems against attacks. This includes adversarial training, differential privacy, encryption, input validation, and other methods to prevent adversarial attacks that could compromise integrity or privacy. By advancing the state of the art in AI security, CoEvolution will directly enhance the EU's defensive capacities against emerging cyber threats targeting AI. Moreover, CoEvolution focuses on standardizing techniques and sharing knowledge across the EU's AI community. Open-source tools, models, and datasets will be provided to catalyze an ecosystem for trustworthy AI aligned with EU values. Collaboration and standardization are explicitly called out as key mechanisms for impact. This fosters harmonized approaches to AI security within the EU. The framework emphasizes regulatory compliance, including with the EU AI Act. This ensures AI systems developed comply with EU regulations and reflect European ethics. It also strengthens oversight and accountability over the use of AI. [**Scale:** *Full AI ecosystem, EU security and standardisation bodies (i.e., AI Act, ELSA, ENISA)* **Significance:** *Ensuring long-term security and reliability of networked AI systems, Dynamically addressing the evolving landscape of security threats and technological advancements in AI cybersecurity domain*]

Targets	Detection rate of adversarial attacks on AI models secured by CoEvolution > 95% (T6.4); Improved robustness of AI models against adversarial examples after applying CoEvolution defences (accuracy degradation <1%, prediction confidence drops) (T4.5); Faster response times in detecting, analyzing and remediating attacks against AI systems secured by CoEvolution by at least 75% (T6.4)
---------	--

Destination Impact #2: More resilient digital infrastructures, systems and processes

CoEvolution proposes to advance the state of the art in adversarial attack detection, prevention, and mitigation. The techniques developed will make **AI models and digital systems built using them more resilient to adversarial threats (Obj1)**. The project also provides tools, techniques, and guidelines for making AI systems compliant with regulations like the EU AI Act, ensuring digital systems integrate regulatory compliance as part of their resilience (**Obj3, 5**). The context awareness mechanisms developed will allow AI systems to dynamically adapt their defences based on the environment, making the systems more resilient to evolving threats. The CoEvolution pilots in complex domains like healthcare will demonstrate the ability of the framework to deliver resilience in mission-critical digital systems and infrastructures (**Obj6**). [**Scale:** *AI engineers, Government Entities, Public Sector and Citizens* **Significance:** *Ensuring safety and trustworthiness in AI applications, Dynamically addressing the evolving landscape of security threats and technological advancements in AI cybersecurity domain*]

Targets	Successful Pilots in Complex Domain = 2 (T6.4); Reduced downtime of mission-critical digital systems and infrastructures enabled by AI models built on the CoEvolution framework by 25-35% (T6.4)
Destination Impact #3: Increased software, hardware and supply chain security, Destination Impact #4: Secured disruptive technologies	
CoEvolution will provide tools and techniques to make AI models more robust and resilient to adversarial attacks. The framework incorporates privacy-preserving measures like differential privacy and encryption to safeguard sensitive data used in training. Continuous monitoring and runtime defenses will secure deployed AI models against emerging threats. The CoEvolution framework accounts also for potential hardware vulnerabilities during model design and training. Defences are built into the model architecture and the context-awareness capabilities allow models to adapt security based on the specific hardware environment during deployment. Hardware-based attacks like model extraction are mitigated through defences like obfuscation. Finally, CoEvolution provides an end-to-end approach covering the entire AI lifecycle from design to deployment. This secures the AI supply chain. The MBOM tracks all components of an AI model, enhancing provenance and accountability. Knowledge sharing, standards, and certification around robust models improves supply chain transparency and trustworthiness. This comprehensive approach taken by CoEvolution, spanning security frameworks, robust models, context awareness, and supply chain security, is expected to significantly advance the resilience and trustworthiness of disruptive AI-enabled technologies (Obj.2,5) . [<i>Scale: Organisations across industries, Government and critical infrastructures, Software/Hardware Vendors, Cybersecurity Industry</i> Significance: Strengthening AI model security, Preserving model integrity and data privacy, Accountability and Responsible AI adoption]	
Targets	Compliance with privacy-preserving measures = 100%; Hardware-based attack mitigation >95% (T6.4); Supply chain transparency and trustworthiness (knowledge sharing, standards, and certification around robust models, indicating enhanced supply chain security) (T2.2, T2.3)
Destination Impact #5: Smart and quantifiable security assurance and certification shared across the EU	
The MBOM approach tracks all components of an AI model and records the security assessment and robustification techniques applied to it. This provides transparency into the model's provenance and security posture. CoEvolution's security assurance engine generates credentials for robust, trusted models that are recorded in the MBOM. These credentials can be verified, updated or revoked as needed over the model's lifecycle. The credentials and MBOM act as a "passport" that provides a transparent, certifiable and verifiable security assurance for AI models shared across the EU . Common standards can be defined for the credentials, thus enabling the harmonized, standards-based certification of AI model security, reliability and trustworthiness as they are shared and deployed across the EU (Obj5). [<i>Scale: Cybersecurity Industry, Regulatory and Standardisation Bodies, End Users of AI Systems, Significance: Providing a comprehensive framework for securing AI models in an AI supply chain, Ensuring compliance with EU directives on trust and fairness</i>]	
Targets	Number of AI models with CoEvolution security credentials ≥ 10 ; Compliance with EU security standards = 100%; Number of AI models with updated security credentials ≥ 10 ; Adoption of AI Model Bill of Materials (MBOM) (T2.4)
Destination Impact #6: Reinforced awareness and a common cyber security management and culture	
Overall, by advancing a comprehensive framework, providing security insights (Obj1,2,4), catalysing standards, ensuring regulatory compliance (Obj3), demonstrating value in pilots (Obj5,6), and driving commercial adoption (Obj7), CoEvolution can significantly reinforce shared awareness, culture, and practices around AI cybersecurity management across the EU . [<i>Scale: Organisations across Industries, AI Researchers and Developers, Cyber-Security Professionals, End Users of AI systems, Significance: Enhancing the security, reliability, and ethical standards of networked AI systems, Offering a holistic AI model security-by-design approach throughout its overall lifecycle and under any AI realization paradigm</i>]	
Targets	Percentage of employees that complete the CoEvolution training programs = 100% (T2.5); Percentage of Security Alerts that are acted upon within defined timeframes >90%; Adoption rate of CoEvolution = 100%; Decreasing risk exposure over time by at least 75% (T6.4)

2.1.3 Economic/Technological Outcomes

From a **technological perspective**, CoEvolution enhances the security, trust, and robustness of AI models throughout their lifecycle. This includes the design phase, the model training/testing/validation phase, and the deployment and operation phase of an AI model. The project introduces a unique data structure, the AI MBOM, which provides detailed information about an AI model, including the dataset used to train it, its dependencies, and all security-related information associated with the use of the CoEvolution solution. CoEvolution also provides a Security Assurance mechanism that records all the assessment and robustification activities an AI model and associated data undergoes in order to become secure/trusted and generates a security credential that accompanies the AI model through its lifecycle. This mechanism also manages the AI model credentials, including credential generation, update,

and revocation.

Targets	Number of robust AI models produced and tested using the CoEvolution framework >5; Attack Detection Rate $\geq 95\%$; Accuracy retention under attack >97%; Attack Prevention Rate~100% (T6.4)
---------	---

From an **economic perspective**, the CoEvolution project can potentially lead to cost savings and increased efficiency in AI model development and deployment. By providing a comprehensive framework for AI model security, CoEvolution can help organizations avoid the costly consequences of security breaches. The project's focus on robustness and trust can also increase the reliability and performance of AI models, leading to improved operational efficiency. Moreover, the CoEvolution project contributes to standard setting in the AI community. The MBOM, for instance, is managed by CoEvolution in an effort to create momentum for its broad adoption by the AI community as a mechanism of reporting robustness, security, trust, explainability, and fairness of an AI model.

Targets	Reduction in costs associated with security incidents or breaches due to the enhanced safety indexes and security metrics; Cost savings from avoided downtime, maintenance, or other related factors; Investments Attracted by businesses or projects using CoEvolution's technologies (T2.4)
---------	---

2.1.4 Societal Outcomes

The societal outcomes and impacts of the CoEvolution project are manifold:

- **Improving Policies and Decision Making:** The CoEvolution project aims to establish a comprehensive risk and security assessment framework for the complete lifecycle of an AI model in a supply chain. This includes the model design phase, the data acquisition/model training phase, and the model deployment and feedback loops. By doing so, it helps to improve policies and decision-making processes related to AI systems, ensuring they are secure, reliable, and ethical.
- **Decreasing CO2 Emissions:** While the CoEvolution project does not directly address CO2 emissions, its focus on improving the efficiency and security of AI systems could indirectly contribute to reducing CO2 emissions. Efficient AI systems can optimize various processes in industries such as energy, transportation, and manufacturing, potentially leading to lower CO2 emissions.
- **Raising Consumer Awareness:** The CoEvolution project aims to provide security, trust, and robustness guarantees for AI models, compliant with EU High-Level Expert Group on AI (AI HLEG) directives on trust and fairness, the European Data Governance ACT, and the GDPR guidelines. This could help raise consumer awareness about the security and trustworthiness of AI applications.
- **Improving AI Supply Chain:** The CoEvolution project aims to provide the foundation for a secure, trusted, and robust AI supply chain ecosystem. This includes secure, trusted, and robust AI models, trusted training datasets, and the framework to utilize them in the AI supply chain.

Targets	Public Awareness and Trust in AI Systems >75% acceptance rate (T2.1); Adoption of CoEvolution's Security Enhancements; Reduction in security incidents or breaches in the AI supply chain; Establishment of the STR AI Supply Chain Ecosystem (T2.4)
---------	--

2.1.5 Feedback to Policy Measures

The CoEvolution project aims to contribute to the design, monitoring, review, and rectification of existing policy and programmatic measures (T2.2), as well as the shaping and support of the implementation of new policy initiatives and decisions by providing real-time context awareness, security enhancement, and robustification throughout the AI model lifecycle, and ensuring the trustworthiness and security of AI systems. The project introduces several mechanisms and tools to achieve these goals, such as the CoEvolution Context Decision Transformer (CDT), Multi-Objective Deep Reinforcement Learning (MODRL) framework, AI Cybersecurity Runtime Monitoring Engine, and AI MBOM Repository. Therefore, feedback to policy measures generated by the CoEvolution project will include the following:

- Utilize the CoEvolution Context Decision Transformer (CDT) to process the generated context vector and define optimal strategies for addressing multi-objective and complex problems in AI model behavior under the presence of an adversary.
- Leverage the Multi-Objective Deep Reinforcement Learning (MODRL) framework to address theoretical and algorithmic impasses that single-objective methods cannot, such as uncertain decisions, limited ability to discover multiple policies, and settings where altruistic agents can see other agents as objectives.
- Implement the AI Cybersecurity Runtime Monitoring Engine to detect anomalies and prevent attacks, such as last layer attacks, GPU overflow, or tainted open-source model deployments.
- Establish the AI MBOM Repository as a critical architecture for ensuring the trustworthiness, resilience, and security of AI systems by providing a central repository for AI model information and facilitating continuous monitoring and evaluation.

2.1.6 Potential Barriers and Requirements

The following table summarizes some of potential barriers that could limit the level of accomplishment of the

project's impacts, and the relevant mitigation actions for reaching the envisaged objectives.

Area of Concern	Requirements/Barriers	Mitigation Actions
Regulatory Environment	The European Union and its member states are making intensive efforts to regulate AI through frameworks like the European Commission's AI Act, aimed at ensuring both safety and trustworthiness in AI applications across all industries.	The project plans to ensure compliance with EU High-Level Expert Group on AI (AI HLEG) directives on trust and fairness, the European Data Governance ACT, and the GDPR guidelines (WP1, T1.5).
Evolving cybersecurity threats	Adversaries will continue developing new attack techniques	Maintain an adaptable framework that can evolve along with the threat landscape (WP3, T3.1).
Engagement with the AI Community	Advancements in frameworks like TensorFlow Federated, ROS, and Horovod, as well as the OpenFog Consortium and Kafka, could influence the CoEvolution project's outcomes	Actively engage with the AI community to promote the adoption of the CoEvolution framework and to gather feedback for further improvements (WP2, T2.1).
Adoption by the AI community	User behavior, including the willingness to adopt new security measures and to change existing practices, could be a significant factor in determining the project's impact	CoEvolution will conduct an ongoing market research to understand the specific needs and challenges of the targeted markets, and adapt the project's approach accordingly (WP2, T2.4).
Targeted Markets	The specific requirements and challenges of targeted markets, as well as their readiness to adopt new AI security measures, could influence the project's outcomes	CoEvolution will implement a robust risk management strategy to identify potential threats to the project's success and to develop appropriate mitigation measures (WP1, T1.3).
Public Scepticism about AI	Reluctance to share data, as well as limited transparency from tech giants, could restrict the development of robust models and tools and the adoption of networked AI innovations	CoEvolution will establish partnerships with academia, industry and policymakers to foster continued R&I in trustworthy AI and will engage with civil society groups that enable AI innovation while protecting public interests and promote awareness and trust in AI. Finally, it will monitor public perceptions and proactively address concerns about networked AI systems via its workshops and info-days (WP2, T2.3).

2.2 Measures to maximise impact - Dissemination, exploitation and communication

The consortium aims to implement an intensive, yet clear, strategy and conduct effective dissemination, communication, and exploitation (D&C&E) activities from the very early stages of the project's lifetime. All partners are committed to mobilize the appropriate stakeholders to multiply the effects of D&C&E activities. Considering the inter-relation between the diverse activities to maximize the project's impact, initially targeted audiences have been identified, as listed in Table

2.2.1 CoEvolution Dissemination plan

The aim of the dissemination strategy is to maximise the outreach of the project for diffusing scientific and technological knowledge to specific target groups (multipliers). Table 2.1 identifies specific dissemination measures and target values, which are tailored to the needs and profile of each target audience, with main aim to gain high visibility and exposure leveraging the multiplier effect through wide means of collaboration.

Table 2.1: Dissemination Plan – Measures, Means, Target Groups and Target Values

Scientific Publications
Scientific Conferences (<i>indicatively</i>): IEEE Int. Conference on Big Data Security on Cloud, Federated Computing Research Conference, IEEE Symposium on Security and Privacy, Computers, Privacy and Data Protection, IEEE Int. Conf. on Cyber Security and Cloud Computing, Symposium on Principles of Database Systems, Privacy Enhancing Technologies Symposium, International Conference on Management of Data, International Joint Conferences on Artificial Intelligence, International Semantic Web Conference, ACM Conference on User Modeling, Adaptation and Personalization, IEEE Scientific Visualization, Computer, Privacy and Data Protection, IEEE Journal of Biomedical and Health Informatics, IoT Week, Digital Around the World, ACM Conference on

Computer and Communications Security. Peer-Review Journals (<i>indicatively</i>): IEEE Transactions on Knowledge & Data Engineering, IEEE Transactions on Big Data, IEEE Transactions on Dependable and Secure Computing, IEEE Transactions on Information Forensics and Security, ACM Transactions on Computer Systems, ACM Transactions of Internet Technology, Journal of Cybersecurity and Privacy, Journal of Machine Learning Research
Target Values: Scientific papers in peer-reviewed Journals: ≥ 10 ; Presentations in Scientific Conferences ≥ 20 .
Technical Publications & White Papers
CoEvolution will publish and contribute to technical blog posts, articles, position/white papers, catalogues, books, or any other reference from technology providers (bottom-up), as well as references related to the use cases under consideration (top-down). Excellent examples of this kind of contribution are the upcoming editions of International Data Spaces, The Data Catalog, Cyber Security Hub White Paper, ICO Blog, and Privacy Matters.
Target Values: Published Technical articles: ≥ 10 ; White papers/factsheets: ≥ 10
Source Code Repository
Encouraging a transparent and open-source culture, CoEvolution will make its software repositories accessible through well-known distributed source code platforms, such as GitHub, GitLab and Bitbucket.
Target Values: Public Deliverables: 10; Open-source code: Publish to 2 repositories (e.g., GitHub, GitLab)
Workshops, webinars, and trainings
Two scientific workshops related to innovative project's technologies will be organised targeting external stakeholders. Also, CoEvolution industrial partners (SIE and TID) will organize internal training sessions to ensure their organization familiarization and further uptake of project's results. Academic partners will also co-organise webinars and training events to PhD schools, as means of knowledge transfer, capacity building, wider outreach, and sustainability.
Target Values: Use Case Workshops: 2; Technical Workshops: 2; Average number of attendees per workshop: >50 Online Training Tutorials: 3; Summer schools: 2; Webinars: 3; Industrial panels: 3; Enrolled users: >500
Push results to Communities, fora, alliances, and associations
Partners will act as liaison to relevant Communities and associations in which consortium partners are active members, to push CoEvolution results. Indicative examples include: BDVA/DAIRO, AIOTI, NGI, NESSI, EurAI), CLAIRE, NetworldEurope, 6G-IA), FIWARE, GAIA-X, Corallia DIH, MultiClouds, ARTIST Open Source community, EU Society for Opinion & Marketing Research.
Target Values: High-impact communities/associations where to push CoEvolution results: >15
Industrial exhibitions, Trade Fairs
Consortium partners will attend well-known symposia, events, industrial exhibitions, and trade fairs to actively interact with multiple stakeholders while disseminating CoEvolution outcomes in the form of presentations, talks, brochures, and personal engagement. Indicative non-exhaustive list of events: EBDVF, EU Data Week, Digital Transformation Conference, AI Summit, Digital Future Congress, IT Future Expo, AI & Big Data Expo, SIDO, EBDVF, Start-up Europe, IoT Week, TCCA Critical Communications World/Europe yearly conferences and exhibitions (CCW), Public Safety Communication Europe (PSCE) conferences. EBDVF, EU Data Week, Digital Transformation Conference, AI Summit, Digital Future Congress, IT Future Expo.
Target Values: Industrial exhibitions targeted: ≥ 10 ; Roll-up banners: 2; Brochures to stakeholders: >1.000
Showcases
One of the main objectives of the project is to showcase and validate the achieved outcomes. CoEvolution will include in its dissemination plan, exhibition/demo activities to demonstrate the feasibility of its proposed capabilities and wide adoption.
Target Values: Showcasing events: ≥ 4 ; Different Countries Involved: ≥ 4 ; Participants: >400 .
Standardisation
To achieve high-scale and long-lasting impact, CoEvolution aims at influencing standards, thus obtaining the seal of approval for its newly implemented components. Interaction with SDOs and other relevant initiatives, such as IEEE, in which consortium partners are active members, as well as contributing to new standards bringing in new requirements to identify and fill gaps, evolving existing standards related to data governance, security and protection, as well as trustworthiness. Such activities are foreseen under task T2.2
Target Values: Number of standardization contributions to different SDOs with adoption status: ≥ 4

2.2.2 Communication plan

The CoEvolution communication strategy and plan focus on creating awareness amongst the full range of

stakeholders including the general public, get high exposure in the social media channels and press coverage, as well as liaise with relevant initiatives for knowledge transfer as outlined in Table 2.2. The aim is to foster the adoption of the project's results by the industry and the society in general.

Table 2.2: Communication Plan – Communication Channels/Target Audience/Timeline and Target Values

Project Website
A project website allowing easy retrieval of main action data with few clicks while acting as the core source of information about the project's objectives and activities. The first version will be online by M01, and will be constantly maintained and updated regularly, running also for 3 years after project end.
Target Values: Online by: M01 ; Visitors at M36: >10.000 .
Social Media
CoEvolution will build and maintain actively its online presence in several social media channels, using X and LinkedIn when interacting with research and innovation communities spreading out new publications, while YouTube and Podcasts will be used for wide community engagement.
Target Values: Social media followers (by M36): >2.000 ; No. of impressions (monthly average): >500 .
e-Newsletters and Email Campaigns
Leveraging the associations and clusters in which partners are members, e-newsletters will provide a snapshot of main activities and achievements of CoEvolution, whereas broadcasting messages to relevant and interested target pools of contact points via email is a highly effective measure of engagement, especially when promoting events and outcome among different domains. Professional marketing platforms (e.g., Sendinblue and Mailchimp) will be used to automate the distribution among contact points
Target Values: e-Newsletters contributed/released: >6 ; Total Mailing List Contact Points: >1.000 .
Online Publishing
Leveraging online publishing platforms (e.g. Medium, WIRED) and magazines (e.g. CORDIS Research EU Magazine) will enable CoEvolution to reach out to a broader audience. Also, all partners will disclose nonconfidential information of the project in their national language to local/regional newspapers, media and press, and the impact of CoEvolution will be promoted. The use case leaders will be leading the spreading and the stakeholder's engagement in the involved countries.
Target Values: Press releases: ≥10 ; Events organised by local authorities: ≥3 ; People per event: >50 .
Links and liaisons
CoEvolution will pursue promotion strategies in various EU-wide initiatives, associations and clusters (e.g. EIT IoT LSP, EIP-SCC, EIP-TEYME) and DIH (e.g., EUHubs4Data, CICERO, TeraLab, CeADAR, DIHSud, Innov:ATE), other ongoing projects (see Section 1.2.5), the relevant CSAs and newly funded projects under CL3 2023-CS-01-03 and to amplify the outreach of the project and to create synergies for promoting the outcomes among the broader community
Target Values: Number of DIH and clusters to contribute: 10 ; Liaisons with other relevant projects: ≥5 .
Promotional material
Promotional material when participating in events, including brochures, catalogues, posters and any other laid out paper-based resource. Most of the material will be available as e-documents and printing will occur as required. CoEvolution will also explore alternatives for labelling gadgets and merchandise, which have turned out quite an effective mean of promoting initiatives among a less specialised audience
Target Values: No. of hard copies (i.e., brochures, flyers, posters, etc.) distributed: >2.000 ; Roll-up Banners: >4 .
Multimedia
CoEvolution will produce multimedia material to have a self-explanatory and appealing presentation of its scope and results, leveraging available distribution channels of promotion (e.g., YouTube, Vimeo). A set of interviews, both from scientific partners and citizens, to collect inputs will be implemented.
Target Values: Number of promotional videos: >3 ; Number of interviews produced: >6 ; views: >3.000
Public Talks
Interact with the general public and inform them about the effect of the CoEvolution results in their everyday life and to create awareness on its societal benefits. This set of activities include the use of social media, online video-clips, public talks at university open days, participation at events organised by consortium partners.
Target Values: Public events at universities: ≥6 ; Students per event: >100 ; Open days: ≥6 ; Visitors/open day: >50

2.2.3 CoEvolution Exploitation plan

The exploitation of the project's results is considered as a key enabler for the success and long-term sustainability of CoEvolution. The CoEvolution consortium has been carefully crafted to embrace complementary expertise and reach out to different domains, as it comprises of industry, SME, research, and academia partners, along with end-users; CoEvolution can therefore establish the necessary relationships with many external players to ensure a widespread promotion and adoption of its results to a wide range of associations, clusters, SDOs, industrial stakeholders and researchers.

As part of task T2.4 "Exploitation activities and long-term sustainability", the consortium will conduct exploitation activities, based on the following exploitation plan:

- **Continuous technology watch and market analysis**, which includes identifying the right potential stakeholders and understanding the challenges that modern industries face in delivering distilling meaningful info from networked AI systems, as well as, listing any gaps in competing and complementary solutions. This entails: (i) a thorough review of the current market status in order to gain insights, capture value-chain trends in the marketplace, identify drivers and/or barriers in the data analytics ecosystem and reach realistic commercial expectations; (ii) a qualitative/quantitative analysis on the economic impact of CoEvolution on how its offering can address emerging market needs and requirements and how they perform against alternatives or competing technological solutions.
- A **joint business model** (possibly through a joint venture) for commercially pursuing CoEvolution beyond the lifetime of the project, which involves, among others, the service types, potential customer segments/collaborators, SWOT analysis, the estimation of costs and pricing schemes, the revenue streams, the definition of a time-plan with appropriate milestones, and the choice of the best channels for reaching the target customer segments. An example could be to offer it on a Platform-as-a-Service (PaaS) subscription-based scheme to vertical players to improve industrial efficiency and consumers' life and work quality.
- The consortium partners **commit to open-source policies** and have decided to take a multi-level approach towards managing the knowledge produced, where the respective knowledge producer holds the rights to the knowledge and shares the software implementation with the rest of the consortium. A key pillar of the CoEvolution exploitation strategy is to offer (post project), free, zero-charge open-source community versions of some of its Key Exploitable Results (KERs) (see **Table 2.3**), to help boost its adoption and therefore solve some of the main issues the data exploitation ecosystem is currently suffering. To support this strategy, CoEvolution will freely deliver online tutorials, webinars and workshops (see Table 2.1) regarding the use and features of its platform and open APIs, so that these can be easily used in a user friendly manner, by (i) interested 3rd party SW developers (SMEs, midcaps, start-ups, etc.), to build, extend and capitalise on CoEvolution platform features, and also adapt and/or evolve them even further to address specific industry needs; (ii) external researchers and future European R&D projects interested to leverage CoEvolution results to cover their specific needs. Such industrial and research 3rd parties will have the opportunity to capitalise on CoEvolution and subsequently provide relevant operational subscription-based and support services on as-a-service or product-based business model to vertical industry players that do not have skills and expertise to operate such a platform.
- The CoEvolution exploitation plan is complemented by (i) the **filing of at least 3 patents** at [EUIPO](#) for protecting the most valuable IP created during the project lifetime; (ii) individual exploitation plans of consortium members, listed in Section 3.2.
- The initial individual exploitation intentions towards the CoEvolution Key Exploitable Results (KER), presented in Table 2.3, which will be **further refined and finalized during the project's duration**.
- All academic and research partners will enrich their scientific knowledge through scientific publications and datasets, broadening their existing knowledge and capacity, creating CoEvolution's scientific knowledge thesaurus and incorporating its scientific innovations in their curricula.

Table 2.3: Project's KER ((Type: Product (P), Service (S), Demonstrators (D)); Open Source (OS); Services: Support (Su), Training (T), Consulting (C), Extension (E)))

Key Exploitable Results	Type	Open Source	Services
CoEvolution Framework	S	Open-source Commercial services	Su, T, C, E
STR Enhancement Framework	P	Yes	Su, T, C, E
STR guarantees for CoEvolution generated AI models	S	Yes	Su, C
Context-awareness mechanism for AI models	P	Yes	Su, T, C, E
AI MBOM	P	Yes	Su, T, C, E

2.2.4 CoEvolution Intellectual Property Management Strategy

Intellectual Property Rights (IPR) Management during the project. Explicit rules concerning IP ownership, access rights to any Background and Results for the execution of the project and the protection of IPRs will be

addressed within the Consortium Agreement (CA), which will be signed by all partners before the project start. **Access Rights to Background and Results.** Explicit rules concerning IP ownership, access rights to any Background and results for the execution of the project and the protection of IPRs will be addressed within the CA. **IP ownership.** Results shall be owned by the partner carrying out the work leading to such results. If any result is created jointly by at least two partners and it is not possible to distinguish between the contribution of each of the partners, such work will be jointly owned by the contributing partners. Any joint results, including inventions and all related patent applications and IP will be jointly owned by the contributing parties.

In order to further the competitiveness of the EU market, and to enhance exploitation of the consortium results, each contributing party to jointly created IP shall have full freedom of action to independently exploit the jointly created results as it wishes (e.g., right to transfer results to their affiliated companies without prior notification to the other project partners), while always protecting and assuring the Access Rights of the other project partners. **Knowledge Management and IPR activities** will be addressed in task **T1.4 ‘Innovation tracking and data management’**. The task will develop and finalise a detailed Knowledge Management and IPR Strategy. An integrated approach will be followed for the identification, capture, retrieval, distribution, sharing, use and reuse of generated information and knowledge assets. This will be essential to drive the value and benefits of these assets and contribute towards maximising the innovation impact of the action. Knowledge management sessions will take place regularly during project meetings and partners will sign compliance declarations of IP policies and procedures.

IP Protection through patents. To protect and retain the commercially strategic IP “for Europe” and “within Europe”, which in turn give exploitation partners a strategic international competitive advantage on a world-wide stage, **at least 3 strategic patent filings** at EU and/or US patent offices are foreseen for protecting the most valuable innovation from the project’s outcomes, in a way that will not impede the underlying aspirations to Open Science. Consequently, the consortium partners have agreed not to publish confidential or restricted results under Open Access, regarding all scientific publications produced along the project lifecycle, until IP filings have been confirmed in the identified strategic and commercially sensitive areas.

2.3 Summary

SPECIFIC NEEDS	EXPECTED RESULTS
▪ Addressing security challenges in collaborative AI environments like decentralized peer-to-peer and federated learning. ▪ Introducing security-by-design, privacy-by-design, and explainability-by-design principles to strengthen AI system defences ▪ Bridging the gap between AI/ML security research and industrial deployment processes. ▪ Ensuring compliance with laws, regulations, and business norms governing AI supply chains.	▪ Development and delivery of CoEvolution novel techniques for adversarial attack detection, prevention, and mitigation in AI Systems ▪ A comprehensive framework for end-to-end security across the AI model lifecycle, ▪ New methods for infusing context awareness capabilities in AI models, and the introduction of the AI Model Bill of Materials (AI MBOM) for tracking an AI model’s components and security posture. ▪ Advanced methodologies for standardized assessment, certification, and continuous monitoring of AI model security ▪ Context-awareness mechanism for AI models and the generated context vectors ▪ AI MBOM principles and DevOps processes ▪ AI Model Management Mechanism ▪ Security, Trust and Robustness Defences
D & E & C MEASURES	
The complete list of actions described in Section 2.2.1, 2.2.2 and 2.2.3 and KPIs included in tables 2.1 & 2.2. Key elements are highlighted below:	
Key Dissemination elements ≥30 scientific publications ≥10 technical articles ≥5 workshops to actively engage with research and innovation communities, ≥10 exhibition/demo activities ≥5 contributions to relevant communities and associations ≥20 participations in industrial exhibitions and trade fairs ≥15 interactions with multiple stakeholders to disseminate CoEvolution outcomes.	Key Standardization elements ≥10 interactions with standardization development organizations, ≥4 contributions to new standards, Key Exploitation elements ≥5 key assets identified, and various business models drafted ≥3 patents filing ≥1 joint business models for commercializing CoEvolution beyond the project's lifetime ≥5 AI communities engaged Key Communication elements website, social media, e-newsletters, email campaigns, online publishing platforms, multimedia material, public talks, and participation in events organized by consortium partners.

TARGET GROUPS	IMPACTS
▪ AI Researchers and Developers ▪ Cybersecurity Industry ▪ Security Professionals and Researchers ▪ Government and Regulatory Bodies ▪ Technology Companies ▪ Software and Hardware Vendors ▪ Industry and Business Leaders ▪ End Users of AI Systems and General Public ▪ Policy Makers	▪ New R&D knowledge to trigger new innovations in the field of Detection, Prevention and Mitigation of Adversarial Attacks on AI models ▪ Improving AI Supply Chain by providing the foundation for a secure, trusted, and robust AI supply chain ecosystem, including secure, trusted, and robust AI models and trusted training datasets. ▪ Improving Policies and Decision Making in AI Systems by establishing a comprehensive risk and security assessment framework for the complete lifecycle of an AI model in a supply chain. ▪ Context Awareness and Resilience: A more resilient and efficient peer-to-peer and federated learning network environment, contributing to the overall robustness of AI systems ▪ Raising Consumer Awareness and trustworthiness of AI applications, promoting security, trust, and robustness guarantees for AI models, compliant with EU directives on trust and fairness, the European Data Governance Act, and GDPR guidelines ▪ Enhanced Security and Trustworthiness: security measures for the model design, training/testing/validation phase, and deployment and operation phase, resulting in AI models with enough associated assurance guarantees to be trusted
OUTCOMES	
▪ Enhanced security and trust in AI models: The CoEvolution project will lead to increased trust in AI models and their applications ▪ Improved resilience against adversarial attacks: The project will help in preventing adversarial attacks that exploit weaknesses in AI algorithms, thereby improving the resilience of AI systems against hostile attacks. ▪ Increased transparency and compliance: The project introduces a new paradigm based on the concept of Bills-of-Materials (BOM) that offers a unified approach to describing AI models in a supply chain. This will facilitate regulatory compliance to the EU approach on Artificial Intelligence such as the EU Artificial Intelligence Act. ▪ Enhanced adaptability of AI models: The project will allow a deployed AI model to be aware of its constantly evolving cybersecurity environment and adapt to adversaries based on such a context. ▪ Creation of a secure AI supply chain ecosystem: The project aims to provide the foundation and create the momentum for a STR AI supply chain ecosystem that offers secure, trusted and robust AI models, trusted training datasets as well as the framework to utilize them in the AI supply chain. ▪ Successful integration and validation of the CoEvolution solution: The project aims to integrate, validate and assess the CoEvolution overall solution in 2 complex pilots that feature all AI model realization paradigms showcasing the security of the overall AI supply chain.	

3 Quality and efficiency of the implementation

3.1 Work plan and resources

3.1.1 Overall structure of the workplan

The purpose of the CoEvolution strategy is to maximise the efficiency of all the activities conducted in the project and the envisioned impact. To this end the structure of the work plan is organised to ensure a synergic cohesion between related tasks but, at the same time, opportunities for concurrency. The project is based on a 36-month plan, and it is divided into 6 work packages (WPs): 4 technical WPs (WP3-WP6), as well as 2 non-technical WPs devoted to Project Coordination (WP1) and Impact maximization (WP2). As such, WP1 provides the necessary activities for the CoEvolution project management. WP3 provides overall technical oversight through state-of-the-art analysis, identification of the most suitable technologies and tools for the purposes of the project, and the requirement elicitation process. It also focuses on the formal definition of the threat models identified in Section 1.2.1.1. This WP will also be in charge of the specification of the overall architecture. The activities of WP3 influence all the other technical WPs (WP4-WP6). WP3&4 are aiming at exploring and providing evidence of the correlation between various parameters and security and privacy risks. At the end, WP4 will also provide tools for mitigating the identified risks, using the identified risk indicators. WP5 provides security assessment and monitoring tools throughout the AI model lifecycle. To this end, it integrates into the AI model design, training and deployment phases, tools and services that provide security assurance, robustness and context-aware adaptation through continuous monitoring engines. WP6 integrates the functionalities developed by WP3-WP5 of the CoEvolution framework, as designed by WP3, within the use cases targeting the demonstration of key features of CoEvolution. Finally, WP2 will communicate

3.1.2 Gantt chart



The CoEvolution project brings together a consortium composed by research centres, universities, industrial companies and SMEs with a proven track record in various facets of the wide spectrum of cybersecurity, privacy, context-awareness, ML robustness and AI systems, a strong commitment towards innovation, and therefore well positioned to efficiently and effectively face the challenges defined in the call. The table below indicates the contribution of the partners in each of those domains that combined will allow to address the project challenges. It is indicative that there are at least 2 partners for each domain, indicating strong complementarity but also leaving room for fault tolerance in case of risk materialization.

The partnership involves fourteen (18) well-established organisations from different countries: Austria, France, Italy, Ireland, Germany, Greece, Cyprus, Switzerland, Spain. The CoEvolution consortium has been brought together based on research excellence, commercial involvement and, most importantly, the complementarity of each partner. The prior collaboration of partners in previous projects and the existence of established business relations and collaborations ensures smooth communication and mutual familiarity with the operational mode of each organization, under a clearly defined collaboration setting.

This set up will enable CoEvolution to achieve its objectives by significantly limiting the risk as multiple partners contribute towards that direction. The following table indicates the how the partners match the project's objectives.

101168560 CoEvolution - Part B - 39

Obj#6					x					x				
Obj#7					x	x	x			x	x	x	x	x

CoEvolution will mobilise resources across the entire consortium to carry out the joint research activities and execute the project in a successful way. Funding will also be allocated to specific initiatives such as the publication of technical outcomes, special journal issues, or attendance to relevant events and standardisation activities as well as marketing and exploitation of project results. All partners have wide experience in EU-funded projects and have gained top-level management approval for this commitment and consequently we can guarantee that the necessary resources will be released to the project.

To further support the complexity of patent research and analysis, with respect to the stated objective of filing at least three patents, or open-source approach, balanced with commercial interests, the consortium plans to mobilize internal resources with relevant expertise. Filing for patents is a common practice for partners like SAG or members of organizations like AVI. Besides experience, those organizations use internal IP councils and/or patent offices to support the filing of patents also with legal support from expert patent writers. Regarding open-source projects, partners like NTUA, UNIPi and UNICA have significant expertise in publishing open-source code and in cases, maintaining open-source projects and supporting communities. We plan to leverage on the experience but also on these communities (e.g. [secml](#), [avalanche](#)) to accelerate the community building process for the CoEvolution open-source projects. Balancing these activities with commercial interests is an open-ended question typically addressed in research projects which we plan to address by setting up rigid processes for defining and implementing an exploitation plan. This plan will reflect the consortiums experience which dictates adaptable schemes in terms of partners participating in an exploitable outcome.

CoEvolution consortium includes different types of institutions and the budget split has been made accordingly, with budget allocated to six (6) SMEs (AEGIS, SLC, AVI, DIE, CBRL, ITML and STS), four (4) large companies (SIE, TID and AVL) and seven (7) research partners (ICCS, ISI, UNICA, IMT, UNIPi, UCY and UCL). Without accounting the indirect costs, the personnel costs are 74% of the total costs, while for other direct costs, the main cost item is composed of travels. For travels, €279,600 have been allocated, considering 9 foreseen project meetings and review and 2-6 trips for dissemination purposes. It is important to highlight that travel for WP-specific meetings will be kept as low as possible by planning back-to-back meetings and using electronic communication, group working tools (e.g., wiki) and audio conferencing. Finally, €30,000 has been reserved for equipment while €41,000 has been allocated for audits.

4 Ethics self-assessment

4.1 Ethical dimension of the objectives, methodology and likely impact

CoEvolution aligns seamlessly with the ethical standards set forth by the EU AI Act, showcasing a commitment to ensuring the safety and trustworthiness of AI applications. The objectives of the research, including the development of a holistic security assessment flow and coherent defenses, resonate with the principles laid out in the EU AI Act. By addressing potential vulnerabilities in AI systems, the research directly contributes to the Act's goal of fostering responsible and trustworthy AI practices across all industries. The GDPR (General Data Protection Regulation) is implicitly acknowledged in the objectives, especially in the consideration of data integrity and the potential societal impacts of compromised AI models. The research's ethical dimension is evident in its alignment with GDPR principles, emphasizing the protection of personal data and privacy within the evolving landscape of AI technologies. Additionally, CoEvolution focuses on enhancing the security and resilience of AI systems by identifying security vulnerabilities and privacy threats across the development and deployment stages. This aims to adopt a security-by-design approach to combat adversarial attacks. However, the challenge lies in locating indicators of these threats within the intricate landscape of single models, collaborative learning architectures and interconnected AI agents/supply chain architectures, encompassing both training and inference phases but also deployment phases. Moreover, an in-depth comprehension of the dynamics of such systems is essential to identify these indicators effectively. Integrating contextual awareness into this pursuit promises more accurate threat detection and robust mitigation strategies, thereby fortifying the overall security posture of those AI systems.

The research methodology demonstrates ethical rigor by considering and incorporating principles from the EU AI Act and GDPR. The proposed unified security assessment framework aligns with the EU AI Act's emphasis on transparency, security, and accountability in AI applications. By adopting a methodology that goes beyond technical aspects to include legal and ethical considerations, the research acknowledges the need for AI security to adhere to regulatory standards. In particular, the methodology of managing the AI Model Bill of Materials (MBOM) throughout the AI model's lifecycle is ethically robust. The concept mirrors the logic of Software Bills of Materials (BOM) used in software solutions, extending this practice to the realm of AI. This approach is not only technically sound but also aligns with ethical considerations by enhancing transparency and accountability. The inclusion of security-related information in the MBOM contributes to the ethical dimension by addressing concerns related to AI model vulnerabilities and potential misuse. The concept of the AI MBOM within the CoEvolution project is closely linked

to several key principles and provisions outlined in the EU AI Act. The EU AI Act, which aims to regulate AI systems within the European Union, provides a legal and ethical framework for the development, deployment, and use of AI technologies as it directly addresses the concepts of:

- Transparency and Information Provision
- Data Governance and Dataset Information
- Security Assurance and Accountability
- Lifecycle Management and Continuous Monitoring
- Community-Wide Adoption and Standards

The GDPR's influence is particularly notable in the research's emphasis on contextual awareness and the responsible handling of data. By addressing biases and ensuring a comprehensive understanding of the operating environment, the methodology reflects a commitment to ethical data practices as mandated by the GDPR. Although it is possible to consider any future issue coming from any possible sort of collection of data, CoEvolution will take care to create a consistent Data Management Plan that will be treated, assessed and structured dynamically, during the whole life-time of the project's effort, so that to "reflect" all potential aspects. DMP will detail the guidelines and methodology used to manage data produced within the project.

CoEvolution will offer practical insights and solutions that align with the regulatory frameworks, promoting ethical AI practices. The impact is not limited to technical advancements but also includes a positive influence on industry practices, ensuring compliance with legal and ethical standards. By incorporating legal and ethical considerations into the proposed security assessment flow, the likely impact extends to the establishment of industry norms that adhere to EU directives and GDPR principles.

4.2 Compliance with ethical principles and relevant legislations

All consortium members are recognised experts and professional legal entities with long prior expertise in EU-funded activities and strong presence in their respective (national, European or other) markets. Due to their professional profiles and their long-presence in the market sector, the members of the consortium exercise their activities so as to be conformant to the legal obligations set by their national frameworks, as well as to be conformant to the respective EU principles and regulation. The consortium members will take specific care to fully conform to any potential requirements coming from ethical principles and/or related national and EU legislation, during the course of the activities.

CoEvolution will ensure respect for the ethical principles and fundamental rights embedded in the regulatory framework of the European Union. These include:

- The European Charter of Human Rights (ECHR)
- The Convention for the Protection of Human Rights and Fundamental Freedoms, especially its arts.7 and 8.
- The EU non-discrimination body of law that ensures no citizen is excluded from essential services on the basis of protected grounds such as gender, race, political or religious beliefs, etc.

The CoEvolution consortium is committed to perform the project in compliance with the General Data Protection Regulation 2016/679 ("GDPR") and any implementing local legislation (collectively referred to as the "EU Data Protection Legislation"). To this end, no Personal Data (as such term is defined in the GDPR) shall be shared by any partner within the consortium, or otherwise provided to another partner, unless (i) it has been fully anonymized prior to the data sharing (thus not technically personal data anymore), or (ii) the specific partners who have elected to exchange or otherwise process Personal Data, have entered into separate agreement (joint controllership, processing or other relevant agreement) and have determined what operational measures should be taken prior to such Personal Data exchange or processing, all in accordance with the EU Data Protection Legislation. Accordingly, in order to ensure compliance with the GDPR provisions, the CoEvolution consortium partners agree to include a representation and warranty clause in the Consortium Agreement that would reflect the aforementioned principles ("Personal Data Clause"). Furthermore, no contractual liability limitation shall apply under the CoEvolution Consortium Agreement to a breach of the GDPR or of the Personal Data Clause, so that, a partner shall be able to fully claim back from a partner which is in breach of the Personal Data Clause.

In terms of AI and Data Governance, it is important to acknowledge frameworks and upcoming legislation on trustworthiness of AI as well as the ethical management of the significant amounts of data used for inference and training of the AI algorithms. CoEvolution consortium is dedicated to ensuring full compliance with the Artificial Intelligence Act (AI act), recognizing its pivotal role in regulating AI technologies within the EU. This commitment extends beyond mere acknowledgment and involves proactive measures to ensure that the CoEvolution project fully aligns with the principles and regulations set forth in the AI Act. As such, the CoEvolution consortium is focused on integrating these principles into the project's fabric. This involves not only adherence to ethical guidelines but also a comprehensive understanding of the Act's regulatory aspects to anticipate and address any potential barriers or challenges that may arise during the development and deployment of the proposed robust STR framework. Furthermore, the CoEvolution consortium recognizes the importance of transparency, accountability, and the

explainability of AI systems, as outlined in the AI Act. As such, we are committed to incorporating mechanisms that ensure these elements are embedded within our software components. Moreover, we aim to engage in continuous monitoring and evaluation of our project's compliance with the AI Act regulations throughout its lifecycle. More specifically, we will consider which legal issues may emerge for the CoEvolution outcomes in the post-project phase, and how the consortium can develop outcomes that are legally compliant and meet a high ethical standard. Then, a second analysis of the legal framework in the second half of the project will be provided, to adapt to changes in the project and to emerging legislations. Finally, we will set up a Legal Requirements Helpdesk to support compliance with all applicable legislation.

In terms of Research Ethics, the project takes into account multiple frameworks to ensure research integrity (i.e. against plagiarism and in-ethical research) as well as the proper scientific experimentation in project PoCs/pilots. The project takes into account:

- The Code of Nuremberg, which includes principles such as properly formulated scientific experimentation
- The Horizon Europe online ethics manual

5 References

- i B. Biggio and F. Roli. Wild patterns: Ten years after the rise of adversarial machine learning. *Pattern Recognition*, 84:317–331, 2018.
- ii A. E. Cinà, K. Grosse, A. Demontis, S. Vascon, W. Zellinger, B. A. Moser, A. Oprea, B. Biggio, M. Pelillo, and F. Roli. Wild patterns reloaded: A survey of machine learning security against training data poisoning. *ACM Comput. Surv.*, 55(13s):294:1–294:39, Jul. 2023
- iii B. Biggio, B. Nelson, and P. Laskov. Poisoning attacks against support vector machines. In J. Langford and J. Pineau, editors, 29th Int'l Conf. on Machine Learning, pages 1807–1814. Omnipress, 2012
- iv T. Gu, B. Dolan-Gavitt, and S. Garg. Badnets: Identifying vulnerabilities in the machine learning model supply chain. In *NIPS Workshop on Machine Learning and Computer Security*, volume abs/1708.06733, 2017
- v B. Biggio, I. Corona, D. Maiorca, B. Nelson, N. Srndic, P. Laskov, G. Giacinto, and F. Roli. Evasion attacks against machine learning at test time. In *ECML PKDD, Part III*, volume 8190 of LNCS, pages 387–402. Springer Berlin Heidelberg, 2013
- vi C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus. Intriguing properties of neural networks. In *International Conference on Learning Representations*, 2014
- vii M. Fredrikson, S. Jha, and T. Ristenpart. Model inversion attacks that exploit confidence information and basic countermeasures. In *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security, CCS '15*, pages 1322–1333, New York, NY, USA, 2015. ACM
- viii B. Sowmiya, VS Abhijith, S Sudersan, R Sakthi Jaya Sundar, M Thangavel, and 1023 P Varalakshmi. A survey on security and privacy issues in contact tracing application of covid-19. *SN computer science*, 2:1–11, 2021.
- ix Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Artificial Intelligence and Statistics, AISTATS'17*, 2017
- x Briland Hitaj, Giuseppe Ateniese, and Fernando Perez-Cruz. Deep models under the gan: Information leakage from collaborative deep learning. In *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security, CCS '17*, 2017
- xi Luca Melis, Congzheng Song, Emiliano De Cristofaro, and Vitaly Shmatikov. Inference attacks against collaborative learning. *arXiv preprint arXiv:1805.04049*, 2018
- xii Xiangru Lian, Ce Zhang, Huan Zhang, Cho-Jui Hsieh, Wei Zhang, and Ji Liu. Can decentralized algorithms outperform centralized algorithms? a case study for decentralized parallel stochastic gradient descent. In *Advances in Neural Information Processing Systems, NIPS'17*, 2017
- xiii Aurélien Bellet, Rachid Guerraoui, Mahsa Taziki, and Marc Tommasi. Personalized and private peer-to-peer machine learning. In *International Conference on Artificial Intelligence and Statistics, AISTATS'18*, 2018
- xiv Fan Mo, Hamed Haddadi, Kleomenis Katevas, Eduard Marin, Diego Perino, and Nicolas Kourtellis. Ppfl: Privacy-preserving federated learning with trusted execution environments. In *Proceedings of the 19th ACM International Conference on Mobile Systems, Applications, and Services, MobiSys'21*, 2021
- xv Muhammad Shayan, Clement Fung, Chris J. M. Yoon, and Ivan Beschastnikh. Biscotti: A ledger for private and secure peer-to-peer machine learning, 2019
- xvi Christodoulos Pappas, Dimitris Chatzopoulos, Spyros Lalas, and Manolis Vavalis. Ipls : A framework for decentralized federated learning, 2021
- xvii Hsin-Pai Cheng, Patrick Yu, Haojing Hu, Syed Zawad, Feng Yan, Shiyu Li, Hai Li, and Yiran Chen. Towards decentralized deep learning with differential privacy. In *Proceedings of Cloud Computing, CLOUD'19*, 2019
- xviii Google Developers. Proximity and cross device communication. <https://developers.google.com/nearby>, 2020
- xix David Silver, Aja Huang, Chris J Maddison, Arthur Guez, George Van Den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, et al. Mastering the game of go with deep neural networks and tree search. *Nature*, 529(7587):484–489, 2016
- xx Hai Nguyen and Hung La. Review of deep reinforcement learning for robot manipulation. In *2019 Third IEEE International Conference on Robotic Computing (IRC)*, pp. 590–595. IEEE, 2019
- xxi Saurabh Gupta, Siddhant Bhambri, Karan Dhingra, Arun Balaji Buduru, and Ponnuram Kumaraguru. Multi-objective reinforcement learning based approach for user-centric power optimization in smart home environments. In *2020 IEEE International Conference on Smart Data Services (SMDS)*, pp. 89–96. IEEE, 2020
- xxii Hao Zheng and Ahmed Louri. An energy-efficient network-on-chip design using reinforcement learning. In *Proceedings of the 56th Annual Design Automation Conference 2019*, pp. 1–6, 2019
- xxiii X. Chen, S. Li, H. Li, S. Jiang, Y. Qi, and L. Song. “Neural model-based reinforcement learning for recommendation,” *CoRR*, vol. abs/1812.10613, 2018
- xxiv Aviv Navon, Aviv Shamsian, Gal Chechik, and Ethan Fetaya. Learning the pareto front with hypernetworks. *arXiv preprint arXiv:2010.04104*, 2020
- xxv Conor F Hayes, Roxana Ra˘dulescu, Eugenio Bargiacchi, Johan Källström, Matthew Macfarlane, Mathieu Reymond, Timothy Verstraeten, Luisa M Zintgraf, Richard Dazeley, Fredrik Heintz, et al. A practical guide to multi-objective reinforcement learning and planning. *Autonomous Agents and Multi-Agent Systems*, 36(1): 1–59, 2022
- xxvi Diederik M. Roijers, Peter Vamplew, Shimon Whiteson, and Richard Dazeley. A survey of multi-objective sequential decision-making. *J. Artif. Intell. Res.*, 48:67–113, 2013
- xxvii Gerald Tesaro, Rajarshi Das, Hoi Chan, Jeffrey O. Kephart, David Levine, Freeman L. Rawson III, and Charles Lefurgy. Managing power consumption and performance of computing systems using reinforcement learning. In John C. Platt, Daphne Koller, Yoram Singer, and Sam T. Roweis, editors, *Advances in Neural Information Processing Systems 20*, Proceedings of the Twenty-First Annual Conference on Neural Information Processing Systems, Vancouver, British Columbia, Canada, December 3–6, 2007, pages 1497–1504. Curran Associates, Inc., 2007
- xxviii Kristof Van Moffaert, Madalina M. Drugan, and Ann Nowé. Scalarized multi-objective reinforcement learning: Novel design techniques. In *Proceedings of the 2013 IEEE Symposium on Adaptive Dynamic Programming and Reinforcement Learning, ADPRL 2013*, IEEE Symposium Series on Computational Intelligence (SSCI), 16–19 April 2013, Singapore, pages 191–199. IEEE, 2013
- xxix Matteo Pirodda, Simone Parisi, and Marcello Restelli. Multi-objective reinforcement learning with continuous pareto frontier approximation. In Blai Bonet and Sven Koenig, editors, *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence*, January 25–30, 2015, Austin, Texas, USA., pages 2928–2934. AAAI Press, 2015
- xxx Kazuyuki Hiraoka, Manabu Yoshida, and Taketoshi Mishima. Parallel reinforcement learning for weighted multi-criteria model with adaptive margin. In Masumi Ishikawa, Kenji Doya, Hiroyuki Miyamoto, and Takeshi Yamakawa, editors, *Neural Information Processing, 14th International Conference, ICONIP 2007, Kitakyushu, Japan, November 13–16, 2007*,

Revised Selected Papers, Part I, volume 4984 of Lecture Notes in Computer Science, pages 487–496. Springer, 2007

^{xxxj} Andrea Castelletti, Francesca Pianosi, and Marcello Restelli. Multi-objective fitted q-iteration: Pareto frontier approximation in one single run. In Proceedings of the IEEE International Conference on Networking, Sensing and Control, ICNSC 2011, Delft, The Netherlands, 11-13 April 2011, pages 260–265. IEEE, 2011

^{xxxii} Pintor, M., Demetrio, L., Sotgiu, A., Melis, M., Demontis, A., & Biggio, B, SecML: A python library for secure and explainable machine learning. arXiv preprint arXiv:1912.10013, 2019.

^{xxxiii} <https://github.com/Trusted-AI/adversarial-robustness-toolbox>

^{xxxiv} Ziwen He, Wei Wang, Jing Dong, Tieniu Tan, Revisiting ensemble adversarial attack, Signal Processing: Image Communication, Volume 107, 2022

^{xxxv} Anuj Dubey, Rosario Cammarota, Vikram Suresh, and Aydin Aysu. 2022. Guarding Machine Learning Hardware Against Physical Side-channel Attacks. J. Emerg. Technol. Comput. Syst. 18, 3, Article 56 (July 2022), 31 pages. <https://doi.org/10.1145/3465377>

^{xxxvi} DENG, Yian; MU, Tingting. Understanding and Improving Ensemble Adversarial Defense. arXiv preprint arXiv:2310.18477, 2023.

^{xxxvii} Casado, F.E., Lema, D., Iglesias, R. et al. Ensemble and continual federated learning for classification tasks. Mach Learn 112, 3413–3453, 2023.

^{xxxviii} Alhassan Mabrouk, Rebeca P. Diaz Redondo, Mohamed Abd Elaziz, Mohammed Kayed, Ensemble Federated Learning: An approach for collaborative pneumonia diagnosis, Applied Soft Computing, Volume 144, 2023

^{xxxix} Papernot, Nicolas, et al. "Distillation as a defense to adversarial perturbations against deep neural networks." 2016 IEEE symposium on security and privacy (SP). IEEE, 2016.

^{xl} Pintor, M., Demetrio, L., Sotgiu, A., Melis, M., Demontis, A., & Biggio, B, SecML: A python library for secure and explainable machine learning. arXiv preprint arXiv:1912.10013, 2019.

^{xli} <https://github.com/Trusted-AI/adversarial-robustness-toolbox>

^{xlii} K. N. Kumar, C. K. Mohan and L. R. Cenkramaddi, "The Impact of Adversarial Attacks on Federated Learning: A Survey," in IEEE Transactions on Pattern Analysis and Machine Intelligence, doi: 10.1109/TPAMI.2023.3322785.

^{xliii} Usman, Muhammad et al. "AntidoteRT: Run-time Detection and Correction of Poison Attacks on Neural Networks." ArXiv abs/2202.01179 (2022): n. pag.

^{xliv} Y. Deng, et al., "An Analysis of Adversarial Attacks and Defenses on Autonomous Driving Models," in 2020 IEEE International Conference on Pervasive Computing and Communications (PerCom), Austin, TX, USA, 2020 pp. 1-10.

^{xlv} B. G. Stottelaar, "Practical cyber-attacks on autonomous vehicles," February 2015. [Online]. Available: <http://essay.utwente.nl/66766/>

^{xlvi} "Researcher hacks self-driving car sensors," <https://spectrum.ieee.org/cars-that-think/transportation/selfdriving/researcher-hacks-selfdriving-car-sensors>, accessed: 2018-08-06.

^{xlvii} National Transportation Safety Board, "Preliminary Report: Highway HWY18MH010," 2018. [Online]. Available: <https://www.ntsb.gov/investigations/AccidentReports/Reports/HWY18MH010-prelim.pdf>.

^{xlviii} A. Unknown, "A Google Self-Driving Car Caused a Crash for the First Time," 2016. [Online]. Available: <https://www.wired.com/2016/02/googles-self-driving-car-may-caused-first-crash>.